HI-TECH BUS

PROJECT REPORT

Submitted in partial fulfillment of the requirement of the award of

BACHELOR OF SCIENCE IN ELECTRONICS

Under the

UNIVERSITY OF CALICUT

By

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Under the Guidance of

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DEPARTMENT OF ELECTRONICS

COLLEGE OF APPLIED SCIENCE, VADAKKENCHERRY

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DEPARTMENTOF ELECTRONICS

CERTIFICATE

This is to certify that the project **"HI-TECH BUS"** is a bonafied work done by **RAKESH.K** submitted in partial fulfillment of the requirement for the award of Bachelor of Science in Electronics under the University of Calicut.

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Date:

DECLARATION

I declare that this submission represents ideas of my own and wherever other Ideas or words have been included , I have adequately cited and referenced the original sources.

I also declare that the project report titled **" HI-TECH BUS"** is a bonafied workdone during the sixth semester B.Sc. Electronics course for the award of Bachelor of Science in Electronics under the University of Calicut.

By

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ABSTRACT

The aim of this project is to develop an automated bus management system that can improve the efficiency and convenience of public transportation. The system utilizes RFID technology and microcontroller boards to track passenger data, including their entry and exit points, travel history, and fare collection. The system also includes a GPS module that helps ensure the accuracy of the data and a digital display that provides real-time information to passengers.

When a passenger scans their RFID tag on the reader, the Arduino Nano microcontroller provides them with destination selection options, and once the passenger selects their destination, the Node MCU microcontroller deducts the travel charge from their user account. The LCD display shows passenger details and balance, and once the servo motor is activated, the passenger can board the bus. The entry count is incremented, and the data is updated to the web server via Node MCU. The GPS module tracks passenger entry and destination, and when the bus reaches the passenger's destination station, the passenger scans their RFID tag on the exit reader, and the RFID details are compared with the web page data. If a match is found, the exit door servo is activated, and the passenger leaves. The exit count is incremented, and the data is updated to the web server via Node MCU.

Overall, this automated bus management system has the potential to revolutionize the way public transportation is managed, making it more streamlined, cost-effective, and convenient for both passengers and bus operators. The system can improve the accuracy of passenger data, reduce the need for cash handling, and enhance the overall commuting experience for millions of people

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CHAPTER 1

INTRODUCTION

In many developing countries, tickets are paper based in most of the public transport system. Public transport system is major attribute for financial gain in any country especially Asian countries. Wireless communication can be defined as transfer of information between two or more points without using wires or cables. There are different wireless technologies such as RFID, IR, GPS, Bluetooth, and WI-FI, etc. In olden days location announcement was done with the help of speakers, but now it is developed by using IVRS in railways stations. Nowadays bus location can be found with the help of Geo Positioning satellites To enable fast and secure, ticket collection we have introduced a hi-tech bus which has an automated ticket machine.

In public transport system a conductor is present in the bus who enquires about the stop .He collects the fare and gives the specific ticket to the passenger .There is no display about the stops in the bus. It is often troublesome because the conductor might not be able to reach all the passengers in the bus. Entering and leaving are done through the same door. The location of the bus cannot be tracked down through any websites. This system can be applied in different areas like transport companies, public trains, private travels, government travel agencies, service organizations, etc. With the advent of GPS and the ubiquitous cellular network, real time vehicle tracking for better transport management has become possible. These technologies can be applied to public transport systems, especially buses, which are not able to adhere to predefined timetables due to reasons

like traffic jams, breakdowns etc. . A Real Time Passenger Information System uses a variety of technologies to track the locations of buses in real time and uses this information to generate predition of bus arrival along the route

1.1 OBJECTIVE

Our aim is to provide a transportation which is convenient and simple. In this system there is an automated ticket machine in the bus which can be used to generate tickets to passengers. The device has the names of all the stops in the specific route. The passenger can select the required stops from the list .The payment can be done via RF tags. There is an LCD display which indicates the current location &the next coming location also how many seats left in the bus. There is also an audio system which alerts the passenger of the current stop. There are two doors-to enter andto exit .The control of the enter door is with the automated ticket machine whereas that of the exit door is with the driver. Moreover ,there is website for the bus which indicates the location of the bus

1.2 BACKGROUND AND MOTIVATION

Background:

Public transportation is an essential part of modern cities, and it is important to ensure that it is efficient and convenient for commuters. However, the current manual methods of collecting bus fares and tracking passenger data can be time-consuming and error-prone. Therefore, there is a need for automated systems that can streamline the process and make it more accurate.

Motivation:

The motivation behind this project is to develop an automated bus management system that can provide a convenient and efficient experience for both passengers and bus operators. By using RFID technology, the system can quickly and accurately track passenger data, including their entry and exit points and their travel history. This data can then be used for various purposes, such as route optimization, fare collection, and passenger analytics. Additionally, the system can reduce the workload of bus operators, freeing them up to focus on other important tasks. Overall, this project has the potential to revolutionize the way public transportation is managed and improve the commuting experience for millions of people.

CHAPTER 2

LITERATURE SURVEY

Tarun Agarwal introduced a project consist of an 8051 microcontroller, a GPSmodem and a LCD display .This entire circuit is placed inside a bus. The power supply provides the power to the entire circuit. The GPS modem consists of a receiver that receive the signals from the satellites based on the latitude longitude and altitude. This coordinate values are stored in microcontroller. These coordinate values are compared with the values from the GPS receiver. These are equal to the latest coordinate values coming from the GPS modem and are converted into TTL level with the help of MAX232. If the latest values match with the stored coordinate values in the microcontroller it displays the location name on the LCD display and announce the stop name by using a speaker IC. This project is helpful to the people who is new to city. [1]

Karthika Rajkumar introduced an AUTOMATIC BUS FARE COLLECTION TECHNOLOGY.The objective of this paper Is to count the passenger using IR sensor and calculating the distance travelled by passenger automatically using motor and u-slot sensor, and the corresponding amount is debited from RFID card. When passenger crosses the signals received will be interrupted and fare collection is done automatically. Here RFID tag is rechargeable one, where as it can be recharged in bus depot or nearest retail shop. [2]

F.Araujo et al discussed the challenge of creating an electronic ticketing system for transportation systems that can partially or completely run on the cloud. This challenge is defined within the scope of an industrial project. The resulting system should be able to reach a large spectrum of customers and should provide two key advantages: lower operational costs, especially for small clients without IT departments, and faster execution of queries for monthly or other sorts of analysis, using the elasticity of cloud-based resources. To fulfill the goals of the project, a system was proposed with very standard technologies and procedures: a threetiered architecture; a separation of the online and analysis databases; and an Enterprise Service Bus to get the input from very diverse hardware and software stacks. In this paper several options regarding the location of these facilities on the cloud was discussed and evaluate the costs involved was evaluated. [3]

BUS TICKET RESERVATION SYSTEM This project intends to computerize its semi computerized ticketing system to provide better customer service. Because of that, The company can provide the easier way of travelling to the Customer or passenger. Electronic tickets, or e tickets, give evidence that their holders have permission to enter a Place of entertainment, use a means of transportation, or have access to some Internet services. Bus Ticket Reservation System enables the bus company's customer to buy bus ticket online.But ,this cannot be used in the case of emergencies as the tickets Should be reserved earlier. [4]

Smart bus management system is based on monitoring the bus Inside bus stand and in order to notify the higher officials about The departure of a bus to the particular place from a lane in time Without any delay. If there is a delay a warning is given using An alarm to a bus driver and the officials regarding that delay. In the daily operation of a bus management system, the Movement of vehicles is affected by uncertain conditions as the Day progresses, such as traffic congestion, unexpected delays, and randomness in passenger demand, irregular vehicle Dispatching times, and incidents. [5]

CHAPTER 3

BLOCK

DIAGRAM

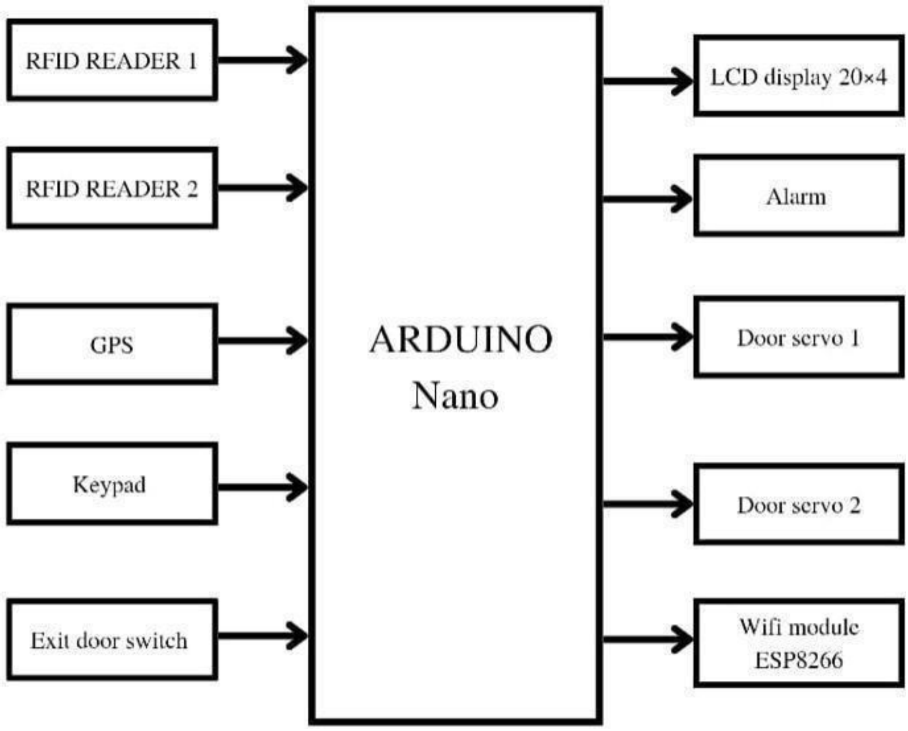


Fig 3.1.Block diagram for hi-tech bus

3.1 Block Diagram Description

Block diagram includes : RFID Readers, Arduino nano ,LCD display, two servo motors GPS, keypad, alarm, wifi module**.** Input is taken through RFID and it processed with the help of LCD display .after signal passed to Arduino then it take action and process the door servo motors which works as door opening and door closing. If the RFID is not valid the arduino does not respond to it and also the door servo does not open.

3.2 RFID READERS

The main input to the system is get by when passenger need verify he need to show his RFID card by showing the card the reader scan and decides whether is it valid or invalid card, this block serves as an reception stage of this entire system. RFID tags contain an integrated circuit and an antenna, which are used to transmit data to the RFID reader (also called an interrogator). The reader then converts the radio waves to a more usable form of data. Information collected from the tags is then transferred through a communications interface to a host computer system, where the data can be stored in a database and analyzed at a later time.

3.3 LCD DISPLAY &3.4KEYPAD

After validating the passenger moves to destination selection process a payment by using the help of keypads and LCD screen the passenger can easily select the destination pay the amount

also with the help of LCD screen we can get information about where the bus currently locating also which is the next station and aditionally how many passenger are inside the bus. The Displaytech 204G series is a lineup of 20x4 character LCD modules. These modules have a 98x60 mm outer dimension with 77x25.2 mm viewing area on the display. The 204G 20x4 LCD displays are available in STN or FSTN LCD modes with or without an LED backlight. The backlight color options include yellow green, white, blue, pure green, or amber color. Get a free quote direct from Displaytech for a 20x4 character LCD display from the 204G series.

3.5 DOOR SERVOMOTOR

The door servomotors are work after the payment which mean the enter door servo motor open the door only after the payment these doors which help full to enter and exit of passengers. SG90 9 g Micro Servo Tiny and light weigh t with high output power.

Servo can rot ate approximately 180 degrees (90 in each direction), and works just like the standard kinds but smaller.

You can use any servo code, hardware or library to control these servos.

Good for beginners who want to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places.

3.6 ALARM

The alarm which is used to alert every station arrival which is help full to give alertto every passenger they never fall sleep.

3.7 WIFI MODULE ESP8266

This unit which provides wifi and also which communicates between arduino also this is connected to gps so all the information of passengers and bus locations and uploded to webserver via this module. This module has a powerful enough on-board processing and storage capability that allows it to be integrated with the sensors and other application specific devices through its GPIOs with minimal development up-front and minimal loading during runtime. Its high degree of on-chip integration allows for minimal external circuitry, including the front-end module, is designed to occupy minimal PCB area. The ESP8266 supports APSD for VoIP applications and Bluetooth co-existance interfaces, it contains a self-calibrated RF allowing it to work under all operating conditions, and requires no external RF parts.

3.8 ARDUINO NANO

The whole unit works under the instruction of Arduino which decides whether door open or not; alarm blow or not transfer passenger details also to node MCU ESP8266. The Arduino NANO is a small Arduino board based on ATmega328P or ATmega628 Microcontroller. The connectivity is the same as the Arduino UNO board. The NANO board is defined as a sustainable, small, consistent, and flexible microcontroller board. It is small insize compared to the UNO board. The Arduino NANO is organized using the Arduino (IDE), which can run on various platforms. Here, IDE stands for Integrated Development Environment. The devices required to start our projects using the Arduino N board are Arduino IDE and mini USB. The Arduino IDE software must be installed on our respected laptop or desktop. The mini USB transfers the code from the computer to the Arduino NANO board.

3.9 GPS

The unit which gives the live location update to the webserve via wifi module. This board features the u-blox NEO-6M GPS module with antenna and built-in EEPROM. This iscompatible with various flight controller boards designed to work with a GPS module.

CHAPTER 4

AIDING DEVICES IN HI-TECH BUS

4.1 ARDUINO NANO

The Arduino Nano is another popular Arduino development board very much similar to the Arduino UNO. They use the same Processor (Atmega328p) and hence they both can share the same program.The Atmel AVR® core combines a rich instruction set with 32 general purpose working registers. All the 32 registers are directly connected to the Arithmetic Logic Unit (ALU), allowing two independent registers to be accessed in a single instruction executed in one clock cycle. The resulting architecture is more code efficient while achieving throughputs up to ten times faster than conventional CISC microcontrollers.The ATmega328/P provides the following features: 32Kbytes of In-System Programmable Flash with Read-While- Write capabilities, 1Kbytes EEPROM, 2Kbytes SRAM, 23 general purpose I/O lines,

32 general purpose working registers, Real Time Counter (RTC), three flexible Timer/Counters with compare modes and PWM, 1 serial programmable USARTs ,1 byte-oriented 2-wire Serial Interface (I2C), a 6- channel 10- bit ADC (8 channels inTQFP and QFN/MLF packages) , a programmable Watchdog Timer with internal Oscillator, an SPI serial port, and six software selectable power saving modes. This allows very fast startup combined with low power consumption.

In Extended Standby mode, both the main oscillator and the asynchronous timer continue to run. Atmel offers the QTouch library for embedding capacitive touch buttons, sliders and wheels functionality into AVR microcontrollers. The patented charge-transfer signal acquisition offers robust sensing and includes fully debounced reporting of touch keys and includes Adjacent Key Suppression® (AKS™)technology for unambiguous detection of key events. The easy-to-use QTouch Suite toolchain allows you to explore, develop and debug your own touch applications. he On-chip ISP Flash allows the program memory to be reprogrammed In-System through an SPI serial interface, by a conventional nonvolatile memory programmer, or by an On-chip Boot program running on the AVR core. The ATmega328/P is supported with a full suite of program and system development tools including: C Compilers, Macro Assemblers, and Program Debugger/Simulators, In Circuit Emulators, and Evaluation kits.

4.1.1 FEATURES OF ATMEGA328

- 28-pin AVR Microcontroller
- Flash Program Memory: 32 Kbytes
- EEPROM Data Memory: 1 Kbytes
- SRAM Data Memory: 2 Kbytes
- I/O Pins: 23
- Timers: Two 8-bit / One 16-bit
- A/D Converter: 10-bit Six Channel
- PWM: Six Channels
- RTC: Yes with Separate Oscillator
- MSSP: SPI and I²C Master and Slave Support
- USART: Yes
- External Oscillator: up to 20MHz

4.1.2 PIN DIAGRAM OF ATMEGA328

Fig 4.1 Pin Diagram of ATMEGA 328P

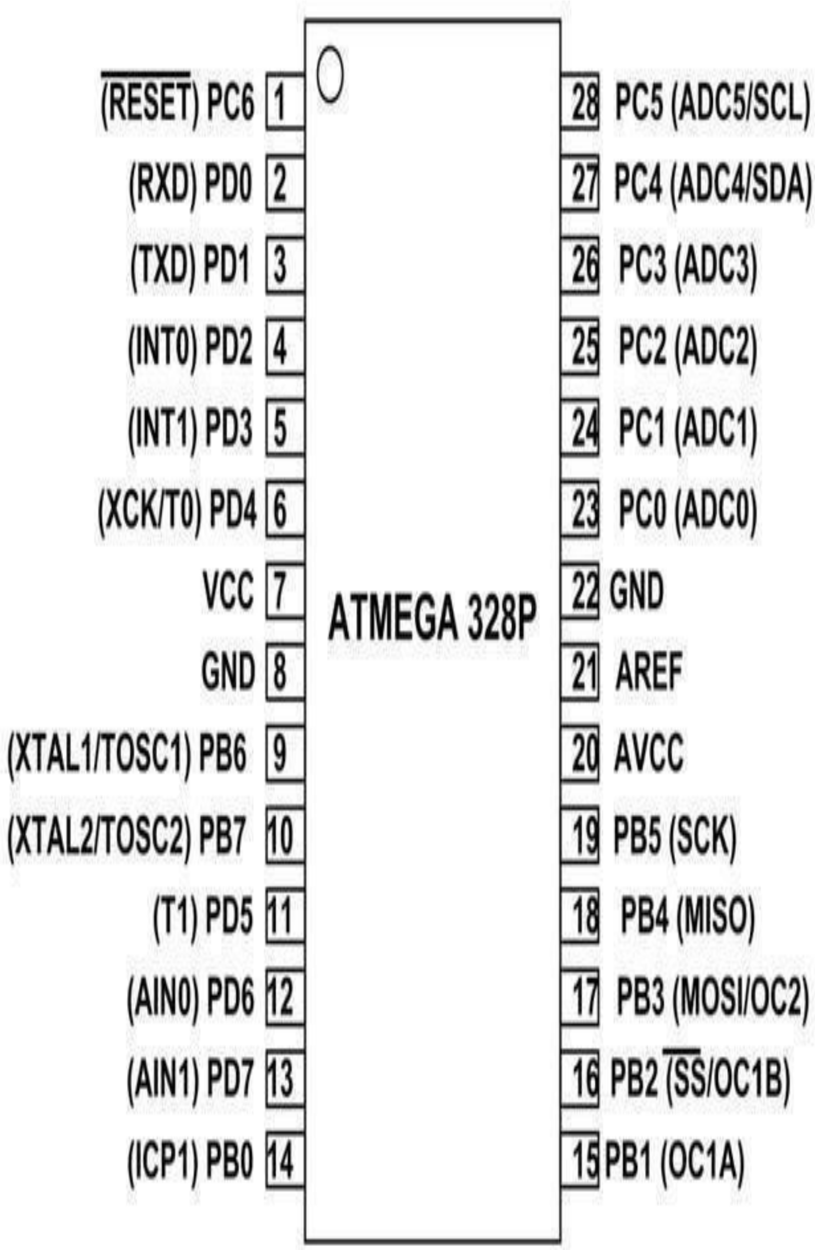


TABLE.4.1.3PIN DESCRIPTION OF ATMEGA328P

PIN NUMBER	DESCRIPTION	FUNCTION
1	PC6	Reset
2	PD0	Digital Pin(RX)
3	PD1	Digital Pin(TX)
4	PD2	Digital Pin
5	PD3	Digital Pin(PWM)
6	PD4	Digital Pin
7	VCC	Positive Voltage(power)
8	GND	Ground
9	XTAL 1	Crystal Oscillator
10	XTAL 2	Crystal Oscillator
11	PD5	Digital Pin(PWM)
12	PD6	Digital Pin(PWM)
13	PD7	Digital Pin
14	PB0	Digital Pin
15	PB1	Digital Pin(PWM)
16	PB2	Digital Pin(PWM)
17	PB3	Digital Pin(PWM)
18	PB4	Digital Pin
19	PB5	Digital Pin
20	AVCC	Positive voltage for ADC
21	AREF	Reference Voltage

22	GND	Ground
23	PC0	Analog input
24	PC1	Analog input
25	PC2	Analog input
26 ,27,28	PC3,PC4,PC5	Analog input

4.1.2 VCC PIN EXPLANATIONS

Digital supply voltage

4.1.2.1 GND

Ground

4.1.2.2 Port B (PB[7:0]) XTAL1/XTAL2/TOSC1/TOSC2

Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port B output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port B pins that are externally pulled low will source current if the pull-up resistors are activated. The Port B pins are tri-stated when a reset condition becomes active, even if the clock is not running.Depending on the clock selection fuse settings, PB6 can be used as input to theinverting Oscillator amplifier and input to the internal clock operating circuit. Depending on the clock selection fuse settings, PB7 can be used as output from the inverting Oscillator amplifier. If the Internal Calibrated RC Oscillator is used as chip clock source, PB[7:6] is used as TOSC[2:1] input for the Asynchronous Timer/Counter2 if the AS2 bit in ASSR is set.

4.1.2.3 Port C (PC[5:0])

Port C is a 7-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The PC[5:0] output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port C pins that are externally pulled low will source current if the pull-up resistors are activated. The Port C pins are tri-stated when a reset condition becomes active, even if the clock is not running

4.1.1.1 PC6/RESET

If the RSTDISBL Fuse is programmed, PC6 is used as an I/O pin. Note that the electrical characteristics of PC6 differ from those of the other pins of Port C. If the RSTDISBL Fuse is un-programmed, PC6 is used as a Reset input. A low level on this pin for longer than the minimum pulse length will generate a Reset, even if the clock is not running. Shorter pulses are not guaranteed to generate a Reset. The various special features of Port Care elaborated in the Alternate Functions of Port C section.

4.1.1.2 Port D (PD[7:0])

Port D is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port D output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs Port D pins that are externally pulled low will source current if the pull-up resistors are activated. The Port D pins are tri-stated when a reset condition becomes active, even if the clock is not running.

4.1.2.4 VCC

AVCC is the supply voltage pin for the A/D Converter, PC[3:0], and PE[3:2]. Itshould be externally connected to VCC, even if the ADC is not used. If the ADC is used, it should be connected to VCC through a low-pass filter. Note that PC[6:4] use digital supply voltage, VCC.

4.1.2.5 VCC

AVCC is the supply voltage pin for the A/D Converter, PC[3:0], and PE[3:2]. Itshould be externally connected to VCC, even if the ADC is not used. If the ADC is used, it should be connected to VCC through a low-pass filter. Note that PC[6:4] use digital supply voltage, VCC.

4.1.2.6 AREF

AREF is the analog reference pin for the A/D Converter.

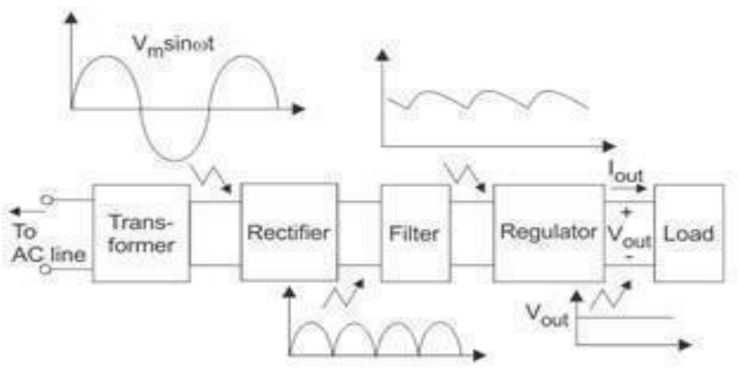
4.1.2.7 ADC [7:6] (TQFP AND VFQFN PACKAGE ONLY)

In the TQFP and VFQFN package, ADC[7:6] serves as analog inputs to the A/D converter.
These pins are powered from the analog suppy and serve as 10-bit ADC channels.

4.1 POWER SUPPLY

In this project we have power supplies with +5V . 7805 is a 5V fixed three-terminal voltage regulator. The IC has internally safe operating features such as short-circuit, over-current, thermal shutdown protection which makes the IC very reliable to make a simple 5V power supply. Must see various type of voltage regulators The first pin of 7805 IC is input, second is Ground while the third one is the Output pin.

BLOCK DIAGRAM OF POWER SUPPLY



Components of typical linear power supply

Fig 4.2 Block diagram of power supply

4.1.3 TRANSFORMER

Transformers are used to convert electricity from one voltage to another with minimal loss of power. They only work with AC (alternating current) because they require a changing magnetic field to be created in their core. Transformers can increasevoltage (step-up) as well as reduce voltage (step-down). Alternating current flowing in the primary (input) coil creates a continually changing magnetic field in the iron core. This field also passes through the secondary (output) coil and the changing strength of the magnetic fi coil is connected to a load the induced voltage will make an induced current flow. The correct term for the induced voltage is 'induced electromotive force' which is usually abbreviated to induced e.m.f.

Eld induces an alternating voltage in the secondary coil.if the secondary

coil is connected to a load the induced voltage will make an induced current flow. The correct term for the induced voltage is 'induced electromotive force' which is usually abbreviated to induced e.m.f.

4.1.4 RECTIFIERS

The purpose of a rectifier is to convert an AC waveform into a DC waveform (OR) Rectifier converts AC current or voltages into DC current or voltage. There are two different rectification circuits, known as 'half-wave' and 'full-wave' rectifiers. Both use components called diodes to convert AC into DC.

4.1.5 FILTERS

A filter circuit is a device which removes the ac component of rectifier output but allows the dc component to the load. The most commonly used filter circuits are capacitor filter, choke input filter and capacitor input filter or pi-filter. We used capacitor filter here.

The capacitor filter circuit is extremely popular because of its low cost, small size,little weight and good characteristics. For small load currents this type of filter is preferred .It is commonly used in transistor radio battery eliminators.

BLOCK DIAGRAM OF CAPACITIVE FILTER

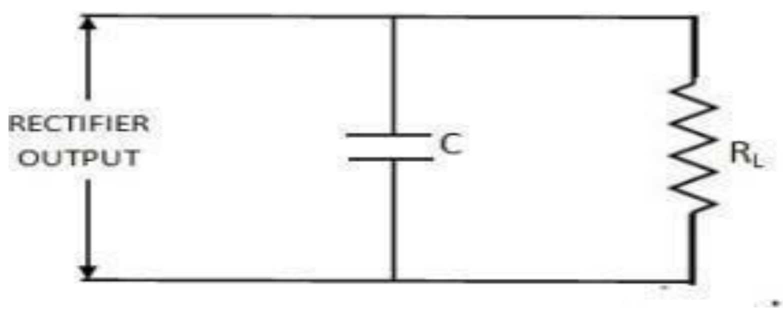


Fig 4.3 Block diagram of capacitive filter

4.2 WiFi Module ESP8266

The ESP8266 is a low-cost Wi-Fi microchip, with built-in TCP/IP networking software and microcontroller capability, produced by Espressif Systems in Shanghai, China.The chip was popularized in the English-speaking maker community in August 2014 via the ESP-01 module, made by a third-party manufacturer Ai-Thinker. This small module allows microcontrollers to connect to a Wi-Fi network and make simple TCP/IP connections using Hayes-style commands. However, at first, there was almost no English-language documentation on the chip and the commands it accepted.The very low price and the fact that there were very few external components on the module, which suggested that it could eventually be very inexpensive in volume, attracted many hackers to explore the module, the chip, and the software on it, as well as to translate the Chinese documentation

Fig 4.3 node mcu

4.2.1 FEATURES OF ESP8266

- Processor: L106 32-bit RISC microprocessor core based on the Tensilica DiamondStandard 106Micro running at 80 or 160 MHz[
- Memory:[6]
- 32 KiB instruction RAM
- 32 KiB instruction cache RAM
- 80 KiB user-data RAM
- 16 KiB ETS system-data RAM
- External QSPI flash: up to 16 MiB is supported (512 KiB to 4 MiB typicallyincluded)
- IEEE 802.11 b/g/n Wi-Fi
- Integrated TR switch, balun, LNA, power amplifier and matching network
- WEP or WPA/WPA2 authentication, or open networks

- 17 GPIO pins[7]
- Serial Peripheral Interface Bus (SPI)
- I²C (software implementation)[8]
- I²S interfaces with DMA (sharing pins with GPIO)
- UART on dedicated pins, plus a transmit-only UART can be enabled on GPIO2
- 10-bit ADC (successive approximation ADC)

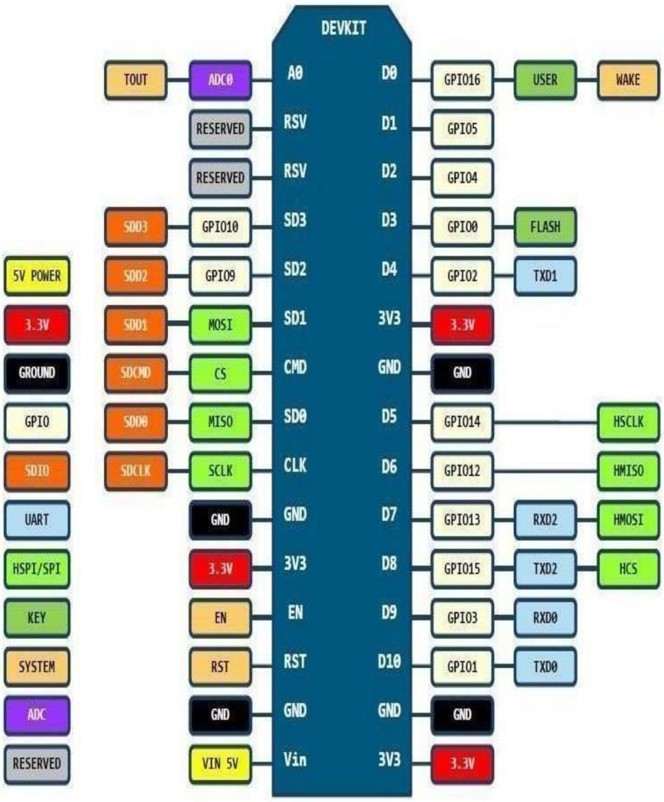


Fig.4.3.1:Pin configuration of ESP826

4.3 GPS

4.3.1 GPS RECEIVER WITH ACTIVE ANTENNA – RS232

Global Positioning System (GPS) satellites broadcast signals from space that GPS receivers, use to provide three-dimensional location (latitude, longitude, and altitude) plus precise time. GPS receivers provides reliable positioning, navigation, and timing services to worldwide users on a continuous basis in all weather, day and night, anywhere on or near the Earth.

Sunroom’s ultra-sensitive GPS receiver can acquire GPS signals from 65 channels of satellites and output position data with high accuracy in extremely challenging environments and under poor signal conditions due to its active antenna and high sensitivity. The GPS receiver's -160dBm tracking sensitivity allows continuous position coverage in nearly all application environments. The output is serial data of 9600 baud rate which is standard NMEA 0183 v3.0 protocol offering industry standard data messages and a command set for easy interface to mapping software and embedded devices.

FEATURES

- · High sensitivity -160dBm
- · Searching up to 65 Channel of satellites
- · LED indicating data output
- · Low power consumption
- · GPS L1 C/A Code
- · Supports NMEA0183 V 3.01 data protocol
- · Real time navigation for location based services
- · Works from +12V DC signal and outputs 9600 bps serial data
- · Magnetic base active antenna with 3 meter wire length for vehicle rooftop

APPLICATIONS

- · Car Navigation and Marine Navigation, Fleet Management
- · Automotive Navigator Tracking, Vehicle Tracking
- · AVL and Location-Based Services
- · Auto Pilot, Personal Navigation or touring devices
- · Tracking devices/systems and Mapping devices application Emergency Locator
- · Geographic Surveying

- · Personal Positioning
- · Sporting and Recreation
- · Embedded applications which needs to be aware of its location on earth PackageIncludes · GPS receiver board (Assembled/Tested with 1 year warranty)
- · Active Antenna with 3 meter cable(Antenna has magnetic base for mounting onvehicletop)
- · Serial Port Cable

4.1.1 GPS INTRODUCTION

The Global Positioning System (GPS) is global navigation satellite systemwhich uses a constellation of between 24 and 32 Medium Earth Orbit satellites that transmit precise microwave signals,that enable GPS receivers to determine their location, speed, direction, and time. GPS has become a widely used aid to navigation worldwide, and a useful tool for map-making, land surveying, commerce, scientific uses, tracking and surveillance, and hobbies such as geo-caching and way marking.Also, the precise time reference is used in many applications including the scientific study of earthquakes and as a time synchronization source for cellular network protocols. GPS has become a mainstay of transportation systems worldwide, providing navigation for aviation, ground, and maritime operations. Disaster relief and emergency services depend upon GPS for location and timing capabilities in their life- saving missions. The accurate timing that GPS provides facilitates everyday activities such as banking, mobile phone operations, and even the control of power grids. Farmers, surveyors, geologists and countless others perform their work more efficiently, safely, economically, and accurately using the free and open GPS signals.

4.3.2 GPS METHOD OF OPERATION

A GPS receiver calculates its position by carefully timing the signals sent by the constellation of GPS satellites high above the Earth. Each satellite continually transmits messages containing the time the message was sent, a precise orbit for the satellite sending the message (the ephemeris), and the general system health and rough orbits of all GPS satellites (the almanac). These signals travel at the speed of light through outer space, and slightly slower through the atmosphere. The receiver uses the arrival time of each message to measure the distance to each satellite thereby establishing that the GPS receiver is approximately on the surfaces of spheres centered at each satellite. The GPS receiver also uses, when appropriate, the knowledge that the GPS receiver is on (if vehicle altitude is known) or near the surface of a sphere centered at the earth center. This information is then used to estimate the position of the GPS receiver as the intersection of sphere surfaces. The resulting coordinates are converted to a more convenient form for the user such as latitude and longitude, or location on a map, then displayed. It might seem that three sphere surfaces would be enough to solve for position, since space has three dimensions. However a fourth

condition is needed for two reasons. One has to do with position and the other is to correct the GPS receiver clock. It turns out that three sphere surfaces usually intersect in two points. Thus a fourth sphere surface is needed to determine which intersection isthe GPS receiver position. For near earth vehicles, this knowledge that it is near earthis sufficient to determine the GPS receiver position since for this case there is only one intersection which is near earth. A fourth sphere surface is also needed to correct the GPS receiver clock. More precise information is needed for this task. An estimate the radius of the sphere is required. Therefore an approximation of the earth altitude or radius of the sphere centered at the satellite must be known.

4. 5 RFID Reader

RFID System is an abbreviation of Radio Frequency Identification System.

It is an "Identification system using wireless communication" that enables transferring data between "RF Tags (or Data Carriers)" that are held by men or attached to objects and "Antenna (or Reader/Writers)". It is a kind of radio communication system.

RFID systems are used in various applications.

Using an RFID system allows consolidated management of objects and information.

Purposes of using RFID in a production site mainly comprises the following applications.

Work instruction (destination instruction)

History management (production history, work history, inspection history, etc.)

Features of RFID Reader:

Able to Read and Write data without direct contact

By "combining an item with its information", a highly pliable and reliable systemconfiguration becomes possible ...

With the adoption of space transmission technology and protocols, highly reliable communication is made possible .

Reading and writing is possible without line of sight, using electric andelectromagnetic wave transmission

4.6 Servo Motor (SG90)

A servomotor (or servo motor) is a rotary actuator or linear actuator that allows for precise control of angular or linear position, velocity and acceleration.[1] It consistsof a suitable motor coupled to a sensor for position feedback. It also requires a relatively sophisticated controller, often a dedicated module designed specifically for use with servomotors Servomotors are not a specific class of motor, although the term servomotor is often used to refer to a motor suitable for use in a closed-loop control system.Servomotors are used in applications such as robotics, CNC machinery, and automated manufacturing.

A servomotor is a closed-loop servomechanism that uses position feedback to control its motion and final position. The input to its control is a signal (either analogue or digital) representing the position commanded for the output shaft.The motor is paired with some type of position encoder to provide position and speed feedback. In the simplest case, only the position is measured. The measured position of the output iscompared to the command position, the external input to the controller. If the output position differs from that required, an error signal is generated which then causes the motor to rotate in either direction, as needed to bring the output shaft to the appropriate position. As the positions approach, the error signal reduces to zero and the motor stops.The very simplest servomotors use position-only sensing via a potentiometer and bang-bang control of their motor; the motor always rotates at full speed (or is stopped). This type of servomotor is not widely used in industrial motion control, but it forms the basis of the simple and cheap servos used for radio-controlled models.More sophisticated servomotors make use of an Absolute Encoder (a type of rotary encoder) to calculate the shafts position, and infer the speed of the output shaft. A variable-speed drive is used to control the motor speed.[3] Both of these enhancements, usually in combination with a PID control algorithm, allow the servomotor to be brought to its commanded position more quickly and more precisely, with less overshooting.

Fig.4.6.sg90servomotor

Features of SG90

- Very small micro size
- 360 degree continuous rotation of the shaft
- Nylon gears
- Analog drive

Unlike normal servos that can be commanded to an exact position within a range of about 180 or 360 degrees, these servos will rotate continuously in either direction with no ability to position it to a particular position. Think of them more like small DC motors with a motor controller built-in that powers the motor and that allows you to start/stop and control direction.The gears are nylon. The nominal speed of rotation is 120RPM at 5V. The motor can be run slower down to about 60RPM, but will stall easier at lower speeds.These types of servos are handy for driving wheels on amicro-robot where space is limited or if you just want to spin something.These are driven by a PWM signal. A pulse width of approximately 1500uS (PWM value of about 185) will cause the motor to stop. A higher PWM value over about 195 will rotate the motor in one direction while a lower PWM value under about 175 will cause the motor to rotate in the opposite direction. If the PWM value drops below about 500uSec (PWM value of 62), the motor will also stop due to insufficient drive

4.7 LCD Display(20*4)

A liquid-crystal display (LCD) is a flat-panel display or other electronically modulated optical device that uses the light-modulating properties of liquid crystals combined with polarizers. Liquid crystals do not emit light directly[1] but instead use abacklight or reflector to produce images in color or monochrome.[2] LCDs are available to display arbitrary images (as in a general-purpose computer display) or fixed images with low information content, which can be displayed or hidden. For instance: preset words, digits, and seven-segment displays, as in a digital clock, are all good examples of devices with these displays. They use the same basic technology, except that arbitrary images are made from a matrix of small pixels, while other displays have larger elements. LCDs can either be normally on (positive) or off (negative), depending on the polarizer arrangement. For example, a character positive LCD with a backlight will have black lettering on a background that is the color of the backlight, and a character negative LCD will have a black background with the letters being of the same color as the backlight. Optical filters are added to white on blue LCDs to give them their characteristic appearance.

The most important feature of this module is that it can display 80 characters at a time.The cursor of this module has 5x8 (40) dots.On this module already assembledthe controller of RW1063.This module operates on the plus five volts input supply andcan also work on the plus three volts.

Fig 4.7.1 :Lcd display(20*4)

4.7.2 Buzzer

The buzzer or beeper is an audio signaling device, which may be mechanical, electro mechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers and confirmation. In the DSI implementation the buzzer is used as an indicator that beeps when the hand is released from fabric and kept idle for more than 30 seconds.

Fig 4.7.2: Buzzer

CHAPTER 5

Circuit Diagram and Working

5.1 Circuit Diagram

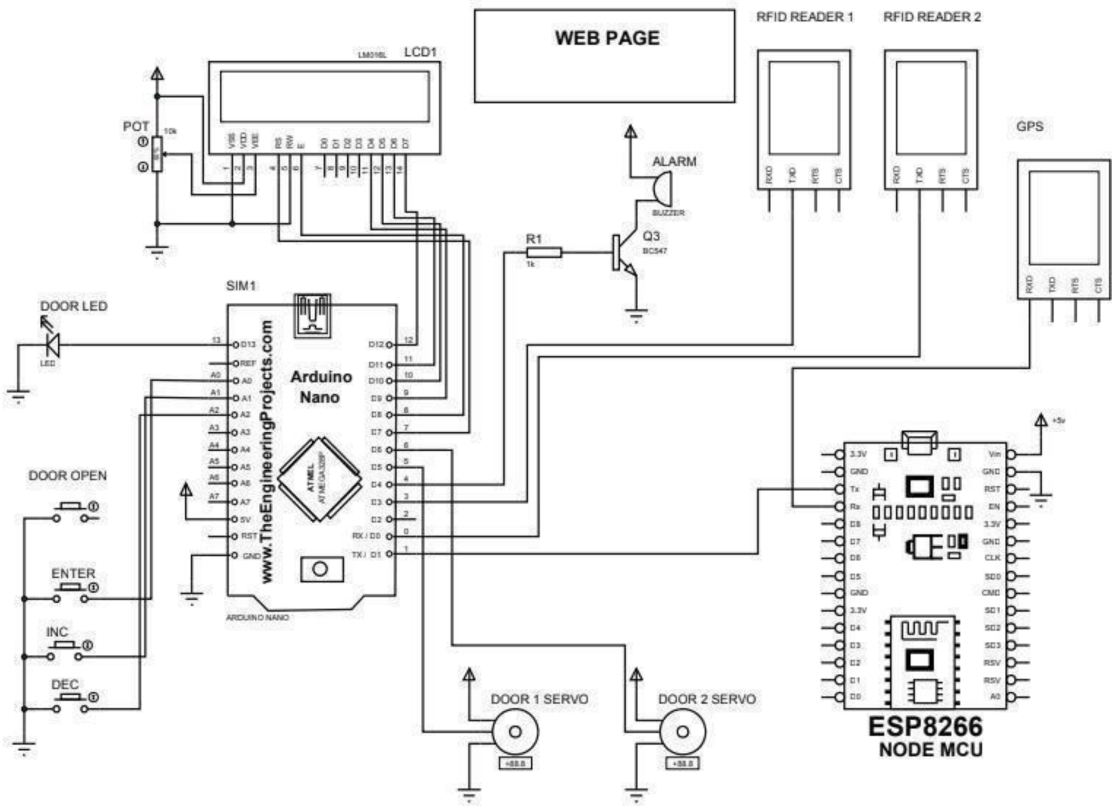


Fig 5.1:circuit diagram

5.2 Working

Basically the whole system consists of audrino nano microcontroller board,a node mcu micro controller board, pair of RFID readers,pair of servo motors ,a digital display,an alarm circuit and an indication light

The whole setup is to be set inside a bus for automated bus management.when apassenger scan his rf id tag on the reader,the audrino nano provide him with the option to select his destination station.after selection the travel charge is cut from his user account via the node mcu.all these details are displayed in the lcd display.after this procedure the servo get activated and the passenger can enter the bus.for every passenger entry the count is incremented and for every passenger who leave,count is decrimented.the same details are updated to the server web via the node mcu controller.the required data for node mcu ois given from the audrino NANO.also the location of the passanger entry and passenger destination is updated to the web with theuse of a gps module.when destination station of a passenger is reached he is required toscan the rf tag at the exit door.rf id details is compared with the details in the web page.When the web data corresponding to the rfid matches. The exit door servo is activated and thus the passenger can leave.the data comparison is made accurate with the help of location access through the gps.

CHAPTER 6
SOFTWARE DESCRIPTION

6.1 Embedded C

Embedded C is a set of language extensions for the C programming language by the C Standards Committee to address commonality issues that exist between C extensions for different embedded systems.Embedded C programming typically requires nonstandard extensions to the C language in order to support enhanced microprocessor features such as fixed-point arithmetic, multiple distinct memory banks,and basic I/O operations. The C Standards Committee produced a Technical Report, most recently revised in 2008 and reviewed in 2013, providing a common standard for all implementations to adhere to. It includes a number of features not available in normal C, such as fixed-point arithmetic, named address spaces and basic I/O hardware addressing. Embedded C uses most of the syntax and semantics of standard C, e.g., main() function, variable definition, datatype declaration, conditional statements (if, switch case), loops (while, for), functions, arrays and strings, structures and union, bit operations, macros, etc.

Let's see the block diagram representation of Embedded C Programming Steps:

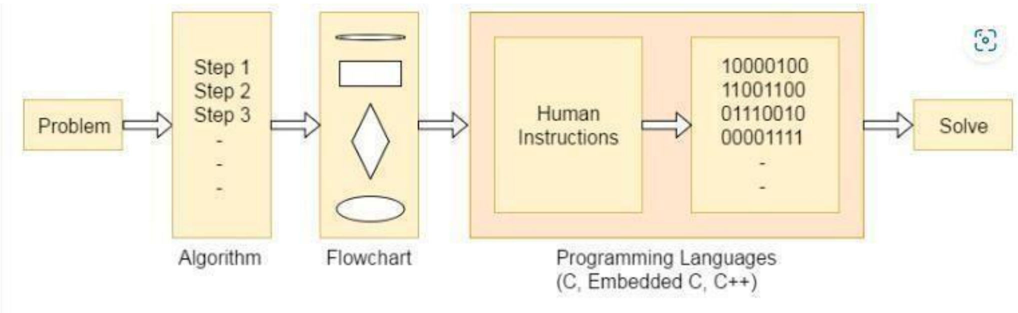


Fig.6.1embedded c

Embedded C Program: Basic

It is important to understand the basics of Embedded C programs.

Embedded C Program: Keywords

You cannot make use of any random words for defining any specific actions because the compiler won't be able to understand the word input provided by you. It is important for the compiler to first identify the word and then perform the specific action assigned (if any). For this to happen there are various predefined words that have a specific meaning defined in accordance with the compiler, known as keywords.

Embedded C Program: Data Types

When we want to declare any variables in the program we make use of data types. Following is the table consisting of all the data types that are used in Keil's Cx51 Compiler.

Data Type Bits(bytes) Range

bit 1 0 or 1 (it is the bit addressable part of RAM)

signed bit 16(2) -32768 to +32767

unsigned bit 16(2) 0 to 65535

signed char 8(1) -128 to +127

unsigned 8(1) 0 to 255

float 32(4) ±1.175494E-38 to ±3.402823E+38

Double 32(4) ±1.175494E-38 to ±3.402823E+

sbit 1 0 or 1 (it is the bit addressable part of

RAM)sfr 8(1) RAM Addresses (considering

80h to FFh) sfr1616(2) 0 to 65535

It is important to have a basic framework in mind before writing any Embedded C program. Following is a practice template that can be used to write your initial Embedded C programs.

Multiline Comments

Single Line

Comments

Preprocessor

DirectivesGlobal

Variables Function

Declarations Main

Function

Local

Variables

Function

Calls Infinite

Loop

Statements

Function

DefinitionsLocal

Variables

Statements

Embedded C Program: Comment

The code of any program should be easily understood by the reader. Sometimes the code of a program can be so lengthy or complicated that it becomes for the developers as well as the reader to make any sense out of it. To eliminate the above stated problems comments are used. Comments are introduced in the code of a program to make it more readable. With the help of comments the reader can understand what is happening in the program. They are mainly used to explain the functioning of the program. The comments are ignored by the compiler, or we can say that they are non executable. A proper documentation of the code can be done with the help of comments for future reference. In complex embedded programming, comments are very useful. Embedded C provides two types of comments: single line comments and multiline comments.

For example:

E=x*y; /* the value of the multiplication of the two variables x and y is stored in another variable E*/

Embedded C Program: Single line Comments

When programming embedded systems there are many fractions of code that require proper explaining for future reference and to increase readability. When you want to only explain a certain fraction of a program it is best to use single line comments. Single line comments begin with (//) double slash, and have to be written ina single line because if the comment is continued in the next line the compiler will not ignore the comment and the program will fail to execute. Single line comments are alsoused to mention the beginning and ending of a certain block of program, promoting a proper structuring of the embedded program.

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For example

E=x-y; /* the value of the subtraction of the two variables x and y is stored in another variable E*/

Embedded C Program: Directives of

ProcessorsAbout Preprocessor Directive

Preprocessing is the first page of the C code compilation phase. The scanning of the provided code and the modifications of the scanned code is done by the preprocessor. The compiler stage receives the code from the processor after the completion of the scanning and modifying process. The commands that are given to thepreprocessor are known as the C Preprocessor directives. Hash symbol (#) is used in the beginning of the preprocessor directives

In the first step the preprocessor scans the complete code and identifies the token # (if any), when the preprocessor identifies a token it modifies the source code, these modifications are done by taking actions that are based on preprocessor directives. When the source code is modified, any preprocessor directives present are eliminated (the directives that begins with the hash symbol (#)). After the termination of the preprocessor directives the code is given to the compiler to complete the process of compilation.

Declaration of new preprocessor directives cannot be done by any programs as the set of directives names (valid) declared are fixed. Before an identifier (which is the name of the directive) hash symbols (#) are used. It is important to notice that there are no semicolons at the end of a preprocessor directive. If you want to declare multi line preprocessor directives then backslashes (\) are used for it. The preprocessor directive used to include files is #include.Before the directive name the symbol hash (#) is used.When including files from standard directory the preprocessor

#include<filename.h> is used. The path of the standard directory is defined in IDE (Integrated Development Environment).

#include is identified as a command by the preprocessor, also the file's content is included in the C source code. Similarly, in embedded C the header files of the microcontroller are included. When the preprocessing process is completed the content from the header files are included in the course code.

6.2 HTML

The HyperText Markup Language or HTML is the standard markup language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript

Web browsers receive HTML documents from a web server or from local storage and render the documents into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for the appearance ofthe document.

HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects such as interactive forms may be embedded into the rendered page. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes, and other items. HTML elements are delineated by tags, written using angle brackets. Tags such as <img /> and <input /> directly introduce content into the page. Other tags such as <p> surround and provide information about document text and may include other tags as sub-elements. Browsers do not display the HTML tags but use them to interpret the content of the page.

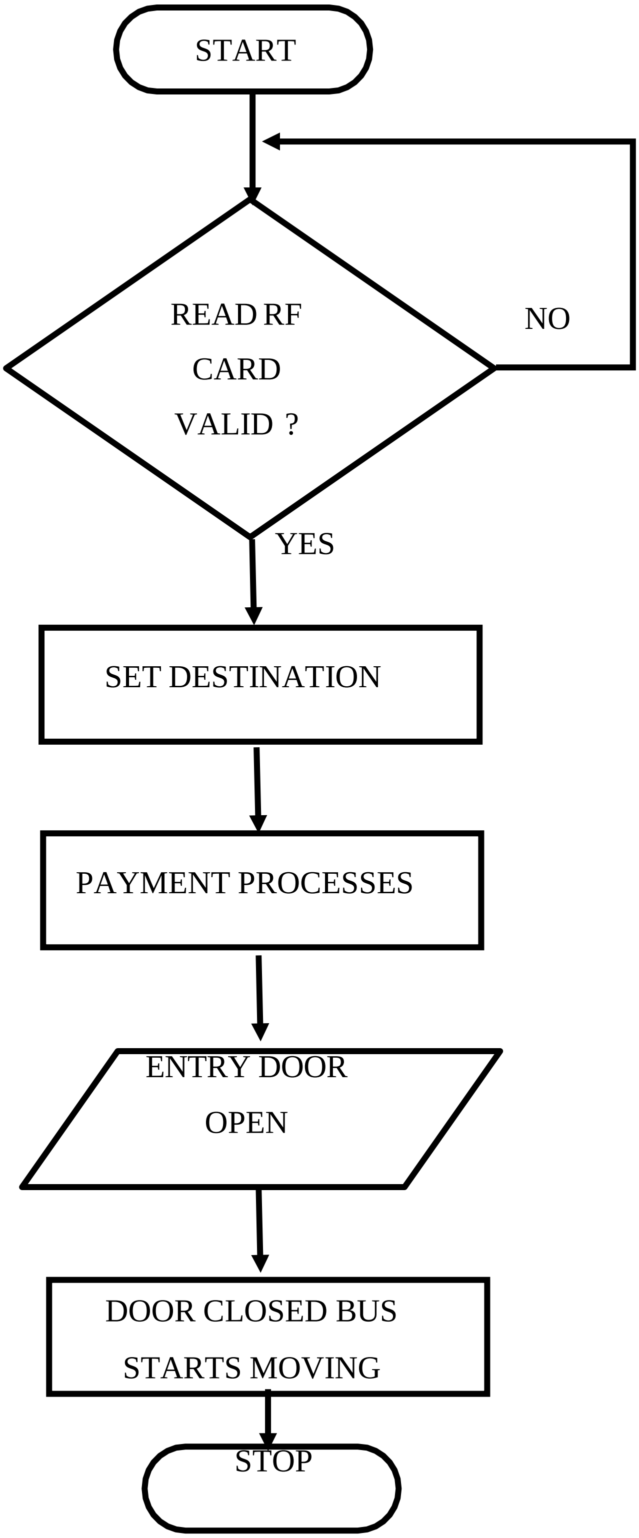
HTML can embed programs written in a scripting language such as JavaScript, which affects the behavior and content of web pages. The inclusion of CSS defines the look and layout of content. The World Wide Web Consortium (W3C), former maintainer of the HTML and current maintainer of the CSS standards, has encouraged the use of CSS over explicit presentational HTML since 1997.[2] A form of HTML, known as HTML5, is used to display video and audio, primarily using the <canvas> element, in collaboration with JavaScript.HTML is an acronym which stands for Hyper Text Markup Language which is used for creating web pages and web applications. Let's see what is meant by Hypertext Markup Language, and Web page.Hyper Text: HyperText simply means "Text within Text." A text has a link within it, is a hypertext. Whenever you click on a link which brings you to a new webpage, you have clicked ona hypertext. HyperText is a way to link two or more web pages (HTML documents) with each other.Markup language: A markup language is a computer language that is used to apply layout and formatting conventions to a text document. Markup language makes text more interactive and dynamic. It can turn text into images, tables, links, etc.Web Page: A web page is a document which is commonly written in HTML an translated by a web browser. A web page can be identified by entering an URL. A Webpage can be of the static or dynamic type. With the help of HTML only, we can create static web pages.

HTML is a markup language which is used for creating attractive web pages with the help of styling, and which looks in a nice format on a web browser. An HTML document is made of many HTML tags and each HTML tag contains different content

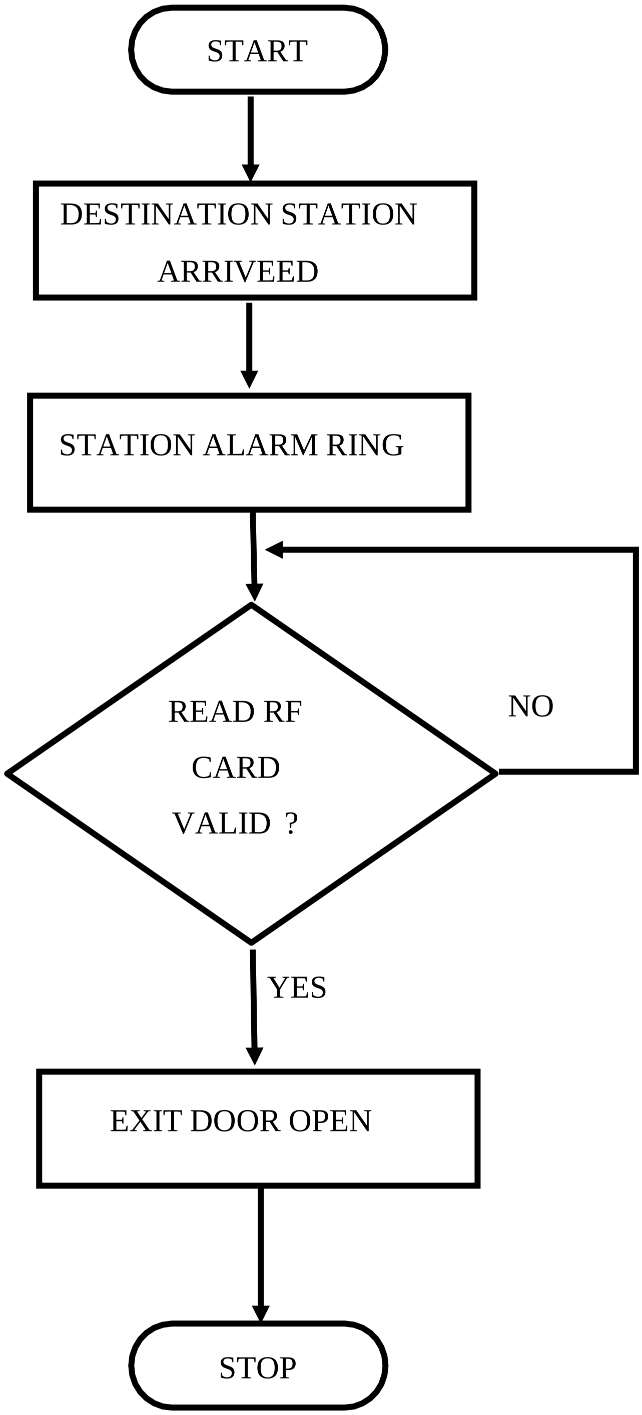
CHAPTER 7

FLOW CHART & PROGRAM

CASE 1.



CASE 2.


START
DESTINATION STATION
ARRIVEED
STATION ALARM RING
READ RF
CARD
VALID ?
NO
YES
EXIT DOOR OPEN
STOP

7.2 PROGRAM

7.2.1 Program of Bus Management

```
#include <LiquidCrystal.h>
LiquidCrystal
lcd(7,8,9,10,11,12);Servo
servo1;
Servo servo2;
 #define alarm         4
 #define door1_servo    5
 #define door2_servo    6
 #define door_led      13
 #define enter_sw      A0
 #define inc_sw        A1
 #define dec_sw        A2
 #define door_sw       A3


unsigned long currentMillis = 0;
unsigned long previousMillis1 = 0;
unsigned int interval = 100,sec=0,sec1=0,msec_100=0,msec1_100=0;

String rfid_1,rfid_2,in_val_1,in_val_2,R_user=" ",validity,data_out="*";
int pos,strt_en=0,Cur_stn=0,Nxt_stn,Exit_en=0,Count=0,dest_en=0,open_en=0;
int                 X_bal=100,Y_bal=100,Z_bal=100,balance,ID=0,destination,fair=10;
//**************************************************************

void door_open();
void
servo_mot_clk();
void
servo_mot_aclk();
void
servo2_mot_aclk();
void setup()
{
  Serial.begin(9600);
  RFID_2.begin(960
  0);servo1.attach(5);
  servo2.attach(6);
  pinMode(enter_sw                    ,
  INPUT_PULLUP);pinMode(inc_sw  ,
  INPUT_PULLUP); pinMode(dec_sw
  , INPUT_PULLUP)
  pinMode(door_sw ,
```

```
INPUT_PULLUP);pinMode(alarm ,
OUTPUT); pinMode(door_led ,
OUTPUT);
digitalWrite(door_led,LOW);
digitalWrite(alarm,LOW);
lcd.begin(20, 4);
lcd.setCursor(0, 0); lcd.print("     WELCOME TO
     ");lcd.setCursor(0,
1); lcd.print(" Bus Management "); lcd.setCursor(0,
2); lcd.print("      System      ");
delay(2000);
lcd.clear();
lcd.setCursor(0, 0);lcd.print("Show ur ID to get
in");lcd.setCursor(0, 1);lcd.print("ID: Bal: ");
lcd.setCursor(0, 2);lcd.print(" Present:    Next: ");
lcd.setCursor(0, 3);lcd.print(" Passenger count: ");
servo1.write(90);
servo2.write(90);
}
void loop()
{
timer();
door_open();
RFID_2_read
();
if( strt_en==0 && digitalRead(enter_sw)==0 ){strt_en=1;delay(100);}
//********************** Stage (1)
if( strt_en==1
&&digitalRead(enter_sw)==0 ){
//********************** Stage(2)
strt_en=2;dest_en=1;
lcd.setCursor(0, 0);lcd.print("Dest:     Fair:
");delay(100);destination=Nxt_stn;
delay(300);
}
if( strt_en==2 && dest_en==1 && digitalRead(enter_sw)==0 )
//********************** Stage (3)
{
delay(300);

lcd.setCursor(17, 1);lcd.print(balance);
if (ID==1){X_bal=balance-
fair;ID=0;Count++;} else if
(ID==2){Y_bal=balance-fair;ID=0;Count++;}else
```

```
                                  if (ID==3){Z_bal=balance-fair;ID=0;Count++;}

                                  lcd.setCursor(0, 0);lcd.print("Success !

                                  ");strt_en=3;dest_en=0;delay(1000);
                                  }
      if( strt_en==3)
      //*************************Stage (4)
          {
           servo_mot_clk();delay(2000);servo_mot_aclk();
           lcd.setCursor(0,     0);lcd.print("Show     ur     ID     to     get-
  in");R_user="";ID=0;strt_en=0;delay(100);
           }
     if ( strt_en==2 &&  dest_en==1 && digitalRead(inc_sw)==0 )
                                 {
                                  destination++; delay(200);
                                  if(destination<=Cur_stn){destination=Nxt_st
                                  n;}lcd.setCursor(5, 0);lcd.print(destination);
                                  fair=(destination-Cur_stn)*10;
                                    if(fair<100){lcd.setCursor(18, 0);lcd.print("  ");}
                                    if(fair<1000){lcd.setCursor(19, 0);lcd.print("
                                  ");}lcd.setCursor(17, 0);lcd.print(fair);
                                  lcd.setCursor(17, 1);lcd.print(balance);

                                }
    if ( strt_en==2 &&  dest_en==1 && digitalRead(dec_sw)==0 )
                                {
                                  destination--; delay(200);
                                  if(destination<=Cur_stn){destination=Nxt_st
                                  n;}lcd.setCursor(5, 0);lcd.print(destination);
                                  fair=(destination-Cur_stn)*10;
                                    if(fair<100){lcd.setCursor(18, 0);lcd.print("  ");}
                                    if(fair<1000){lcd.setCursor(19, 0);lcd.print("
                                  ");}lcd.setCursor(17, 0);lcd.print(fair);
                                  lcd.setCursor(17, 1);lcd.print(balance);
                                 }

   if( strt_en==0
             { lcd.setCursor(0, 0);lcd.print("Show ur ID to get-in");lcd.setCursor(3,
  1);lcd.print("    ");RFID_1_read();}
   if (strt_en==1 && dest_en==0)
              { //lcd.setCursor(0, 0);lcd.print("Show ur ID to get
               in");lcd.setCursor(0, 0);lcd.print("Set destination  ");
               if(balance<10){lcd.setCursor(18, 1);lcd.print(" ");}
               if(balance<100){lcd.setCursor(19, 1);lcd.print(" ");}
```

```
        lcd.setCursor(17, 1);lcd.print(balance);
       }
 Nxt_stn=Cur_stn+
       1;
 lcd.setCursor(9, 2);lcd.print(Cur_stn);lcd.setCursor(18,
2);lcd.print(Nxt_stn);if(Count<=0){Count=0;} lcd.setCursor(17,
3);lcd.print(Count);

// if(balance<10){lcd.setCursor(18, 0);lcd.print("  ");}
 //if(balance<100){lcd.setCursor(19, 0);lcd.print(" ");}



 if
    (ID==1){balance=X_bal;
 } else
 if(ID==2){balance=Y_bal;}
 else
 if(ID==3){balance=Z_bal;}
 else{lcd.setCursor(17, 1);lcd.print("  ");}

//     Serial.print("strt_en:   ");   Serial.print(strt_en);Serial.print("   Dest_en:
       ");Serial.print(dest_en);
// Serial.print(" ID: "); Serial.println(ID);

if(sec1>5){data_out=data_out+Count+Cur_stn;Serial.println(data_out);delay(100);data
_out="*";sec1=0;}
}


void door_open()
{

 if(                      open_en==0
                          &&digitalRead(door_sw)==0
){open_en=1;sec=0;Cur_stn++;delay(300);} if(open_en==1){
digitalWrite(door_led,HIGH);delay(200);}
 if(open_en==1 && sec>15){open_en=0;digitalWrite(door_led,LOW);delay(200);}
void RFID_1_read()
{
  if (Serial.available() > 0)
  {
   rfid_1 = Serial.readString();delay(25);
   in_val_1=rfid_1.substring(0,12);
   Serial.print("rfid_1:");Serial.println(in_val_
```

```
    1);
     if              (in_val_1=="870081356754"){R_user="User-
X";ID=1;validity="Valid";strt_en=1;delay(100);} // 870081356754X
     else   if    (in_val_1=="87007F342DE1"){R_user="User-
           Y";ID=2;validity="Valid";strt_en=1;delay(100);}   // 87007F342DE1        Y
     else    if    (in_val_1=="8700858636B2"){R_user="User-
           Z";ID=3;validity="Valid";strt_en=1;delay(100);}   // 8700858636B2        Z
     else                                                          {R_user="    ";
ID=0;validity="Invalid";strt_en=0;delay(100);}
 //  Serial.print(" Matched user_1: ");Serial.println(R_user);
 //  lcd.setCursor(3, 1); lcd.print(validity);delay(250);
    //lcd.setCursor(14, 1);lcd.print("       ");
    //Serial.print(" Matched ID: ");Serial.println(R_ID);lcd.setCursor(14,
1);lcd.print("");
  }
}
void RFID_2_read()
{
   if (RFID_2.available() > 0)
   {
    rfid_2 = RFID_2.readString();delay(25);
    in_val_2=rfid_2.substring(0,12);
    Serial.print("rfid_2:");Serial.println(in_val_
    2);
     if      (open_en==1 && in_val_2=="870081356754")
           {R_user="User-
X";servo2_mot_clk();delay(2000);servo2_mot_aclk();open_en=0;
                                                         Count--;
delay(100);digitalWrite(door_led,LOW);}   // 870081356754     X
     else if (open_en==1 && in_val_2=="87007F342DE1")
           {R_user="User-
Y";servo2_mot_clk();delay(2000);servo2_mot_aclk();open_en=0;
                                                          Count--;
delay(100);digitalWrite(door_led,LOW);}   // 87007F342DE1     Y
     else if (open_en==1 && in_val_2=="8700858636B2")
           {R_user="User-
Z";servo2_mot_clk();delay(2000);servo2_mot_aclk();open_en=0;
                                                                   Count
--;delay(100);digitalWrite(door_led,LOW);}  // 8700858636B2

  }
}
void timer()
{
  currentMillis = millis();
```

```
 if ((unsigned long)(currentMillis - previousMillis1) >= interval) // check for rollover
delay 1

   msec_100++;msec1_100++;

}
//*********************************************************************
**
void servo_mot_clk()
{
 Serial.println("Gate   opening   ");
 for (pos = 90; pos <= 180; pos +=
 2)
 {
   digitalWrite(alarm,HIG
   H);servo1.write(pos);
   delay(30);
 }
 digitalWrite(alarm,LOW);
}
void servo_mot_aclk()
{
 Serial.println("Gate   closing   ");
 for (pos = 180; pos >= 90; pos -=
 2)
 {
  digitalWrite(alarm,HIG
  H);servo1.write(pos);
  delay(30);
 }
 digitalWrite(alarm,LOW);
}
//*********************************************************************
**
void servo2_mot_clk()
{
 Serial.println("Gate opening

 ");for (pos = 90; pos <= 180;
 pos +=
 {
   digitalWrite(alarm,HIG
   H);servo2.write(pos);
   delay(30);
 }
```

```
  digitalWrite(alarm,LOW);
}
```

7.2.2 Website Program

```
//D1-CLK,D2-DATA FOR I2C LCD
#include
<ESP8266WiFi.h>
#include <WiFiClient.h>
#include
<ESP8266WebServer.h>
#include <ESP8266mDNS.h>
#include <Wire.h>
#include
<LiquidCrystal_I2C.h>
#include <TinyGPS++.h>
TinyGPSPlus gps;
#include
 <SoftwareSerial.h>
 #define rxPin D7
 #define txPin D8
 SoftwareSerial serial_2(rxPin,txPin);

ESP8266WebServer server(80);
LiquidCrystal_I2C lcd(0x27,16,2);


String
lattitude,longtitude,URL_location,URL_send,loc ;
const char* ssid = "Bus management";
const char* password = "12345678";

unsigned long currentMillis = 0;
unsigned long previousMillis1 = 0;
unsigned int interval =
100,sec=0,msec_100=0,msec1_100=0;unsigned char
flag=0,alert=0;
int
Cur,Nxt,Count,Seats,total=20;
String data,in_val;
void handleRoot()
{
  String cm
```

```
  cmd += "<!DOCTYPE HTML>\r\n";
  cmd +=
  "<html>\r\n";cmd
  += "<head>";

  cmd += "<title>\"CONTROL PANEL\"</title>";
  cmd += "<meta http-equiv='refresh'
  content='3'/>";

  cmd +=
  "<style>";cmd
  += "a {";
  cmd += " width:200px;";
  cmd += "display: block;";
  cmd += "border-radius:10px;";
  cmd += "background-color:
  grey;";cmd += "border: red;";
  cmd += "color: black;";
  cmd += "padding: 7px ;";
  cmd += "text-align: center;";
  cmd += "text-decoration: none;}";
  cmd += "a:hover {background-color: yellow;}";//hower colour
  selectioncmd += "</style>";
  cmd += "</head>";

  cmd += "<h1> MONITORING UNIT</h1>";
  cmd += "<h2> BUS MANAGEMENT SYSTEM</h2>";

  cmd += "<h3>Location-----:  " + loc +  "</h3>";
  cmd += "<h3>Present station: " + String(Cur) + "</h3>";
  cmd += "<h3>Next station:     " + String(Nxt) + "</h3>";
  cmd += "<h3>Passenger count: " + String(Count) +
  "</h3>";cmd += "<h3>Seats available: " + String(Seats) +
  "</h3>"; cmd += "<html>\r\n";
  server.send(200, "text/html", cmd);
}

void handleNotFound()
{
  String message = "File Not
  Found\n\n";message += "URI: ";
  message += server.uri();
  message += "\nMethod: ";
  message += (server.method() == HTTP_GET)?"GET":"POST";
```

```
  message += "\nArguments:
  ";message += server.args();
  message += "\n";
  for (uint8_t i=0; i<server.args(); i++)
  {
    message += " " + server.argName(i) + ": " + server.arg(i) + "\n";
  }
  server.send(404, "text/plain", message);
}
void timer();
void serial_2_read();
void
GPS_LOC_TAKE();
void setup(void)
{

  Serial.begin(9600);
  serial_2.begin(960
  0);

  delay(2000);
  WiFi.mode(WIFI_AP_STA); //need both to serve the webpage and take
commandsvia tcp
  Serial.println("");
  IPAddress
  ip(1,2,3,4);
  IPAddress gateway(1,2,3,1);
  IPAddress subnet(255,255,255,0);
  WiFi.softAPConfig(ip, gateway, subnet);
  WiFi.softAP(ssid,password); //Access point is open

  delay(1000);
  IPAddress myIP =
  WiFi.softAPIP();Serial.print("AP
  IP address: ");
  Serial.println(myIP);

  if (MDNS.begin("esp8266")) { Serial.println("MDNS responder started");
  }server.on("/", handleRoot);

  server.onNotFound(handleNotFound);

  server.begin();
  Serial.println("HTTP server started");
```

```
}

void loop(void)
{
  serial_2_read
  ();timer();
  if(msec1_100>2){server.handleClient();msec1_100=0
  ; }Seats=total-Count; Nxt=Cur+1;
  if(sec>10){flag=2
   ;}if(flag>1)
    {
     Serial.println("http://www.google.com/maps/place/
      ");delay(500);
      while (Serial.available() > 0)
      {
        if (gps.encode(Serial.read()))
        {
          flag++;
          if(flag>
          7)
          GPS_LOC_TAKE();
          if(alert==1)
          {
            delay(1000);
            loc=URL_locatio
            n;alert=0; flag=0;
            delay(500);break
            ;
          }
        }
        yield() ;
      }
    }
}
void timer()
{
  currentMillis = millis();
  if ((unsigned long)(currentMillis - previousMillis1) >= interval) // check for rollover
delay 1
    msec_100++;msec1_100++;
    if(msec_100>9){msec_100=0;sec++
    ;}previousMillis1 = currentMillis;
  }
}
void serial_2_read()
```

```
{
  if (serial_2.available() > 0) //*C#S
  {
   data = serial_2.readString();delay(25);
  //
   Serial.print("data:");Serial.println(in_val)
   ;String f_val=data.substring(0,1);
   if(f_val=="*")
   {
   String
   data1=data.substring(1,2);
   String
   data2=data.substring(2,3);
   Count=data1.toInt();
   Cur=data2.toInt();
   }
  }
}

void GPS_LOC_TAKE()
{
 URL_location =
 "http://www.google.com/maps/place/";if
 (gps.location.isValid())
 {
   lattitude = String(gps.location.lat(), 6);
   longtitude = String(gps.location.lng(), 6);
   URL_location = URL_location +
   lattitude; URL_location = URL_location
   + ","; URL_location = URL_location +
   longtitude;
   Serial.print("Location URL: ");
   Serial.print(URL_location);delay(1000);
   alert=1;
 }
 else{URL_location=URL_location+"10.592517,76.467427";alert=1;}
}
//10.592517, 76.467427
```

CHAPTER- 8

IMPLEMENTATION AND RESULT

The ‘HiTech Bus’ system is implemented successfully .They are developed and tested.The implementation of the project is initiated by selecting hardware to implement the system, this begins with choosing suitable microcontrollers.The major controllers are the Arduino NANO and Nodel MCU ESP8266.

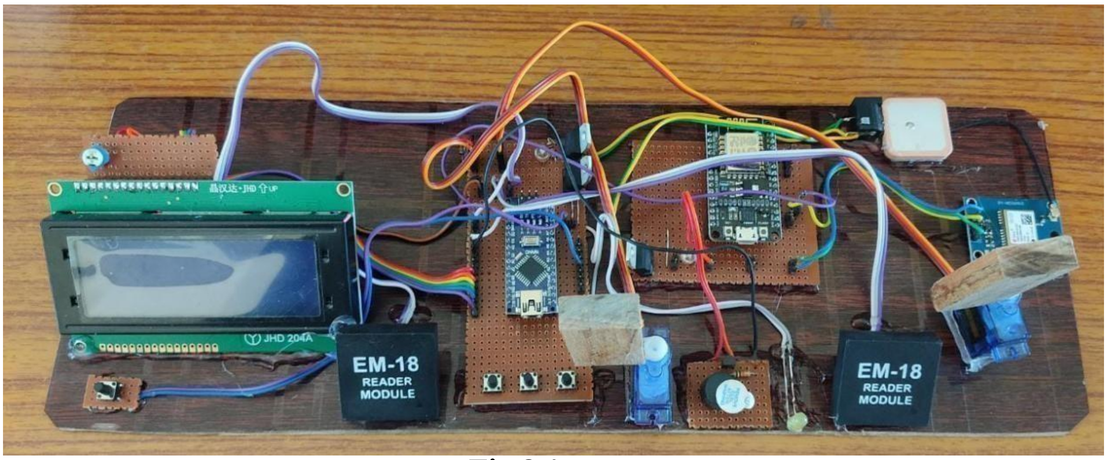


Fig 8.1 prototype

During the first stage, we have completed the RFID section of our project.It consist of Arduino NANO, two RFID Readers, two Servo motors as two doors.RFID Reads the RF card and send instructions to the Arduino NANO.After RF ID verification the passenger can select destination location through using keypad and lcd display.After payment for the needed location, the doors will be open and the passengers can enter into the bus. While the bus moving,current location of the bus and the passenger count in the bus can be known by using website.The buzzer will announce every location So the passenger can leave the bus by verifying the RF ID then the doors will be open.

CHAPTER-9

ADVANTAGES AND LIMITATIONS

9.1 ADVANTAGES

Manual effort can be reduced since there is no conductor required.It provides fast and secure ticket collection system. Since the website provides the information of the bus, it is easy to track down the bus. Also in this bus the exit door control was in drivers so we can easy to catch the spam passenger

The paymet for required destination is done through the RFID; so we does not need an external man as fair collector another advantage of this system was we can get all updates of passengers count and current station through oru webserver via wifi module NODE MCU

By implementing this system we does not need to keep liquid money always weonly keep the RF card to travel anywhere this complete unit works on the dc adapter power we no need of battery avoids recharging time.

The complete unit places in different part of the bus also the paymenting unit isless in size

9.2 LIMITATIONS

Sometimes,the GPS might not be able to track the exact location .

CHAPTER-10

CONCLUSION AND FUTURE SCOPE

CONCLUSION

In our project,we have developed a system which has an automated ticket machine which collects the fare and provides the ticket to the passenger. Only need to keep the RF card for travelling , hence we are providing advanced passenger facilities like station alarm which is very help full for sleepy passengers which gives alert the automated bus management system presented in this project is a highly efficient and reliable solution for managing passenger entries and exits on a bus. The use of RFID technology, servo motors, a digital display, an alarm circuit, and an indication light, along with the Arduino Nano and Node MCU microcontrollers, enable the system to operate smoothly and accurately. By integrating GPS location access and web data updating, the system provides real-time data on passenger entries and exits, making it a valuable tool for bus operators and passengers alike. Overall, this project demonstrates the potential of advanced technology for improving public transportation and creating a more seamless and convenient travel experience.

The main goal of this project is to provide advanced technologies with easy way of transportation.

FUTURE SCOPE

1. provide station name announcement with 8-channel APR MODULE.
2. Launch an official application for all updates and information collecting of everypassengers.
3. also providing an touch screen display by replacing of LCD display.4.providing an GSP route map screen infront of driver
5. update the webserver to an official website where we can track the bus locationupdates instantly
6. update the GPS unit for faster and accurate location updates

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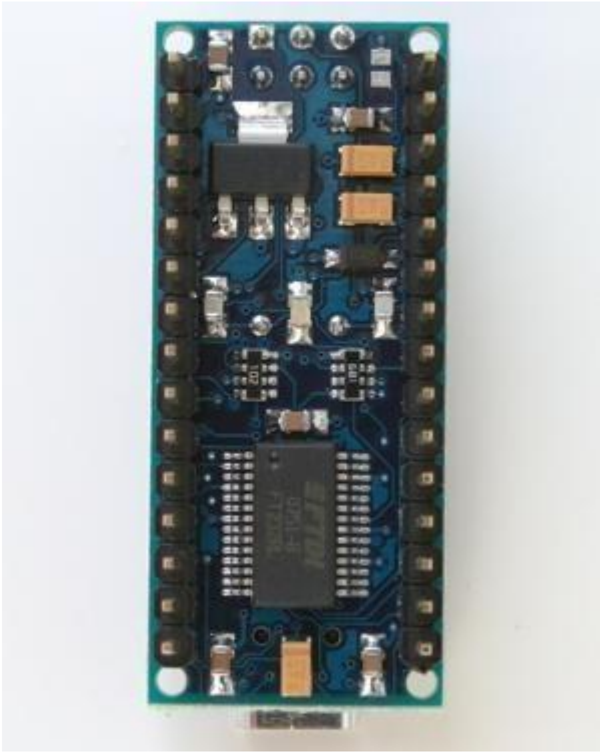
APPENDIX

Arduino Nano

Arduino Nano Front



Arduino Nano Rear

Overview

The Arduino Nano is a small, complete, and breadboard-friendly board based on the ATmega328 (Arduino Nano 3.0) or ATmega168 (Arduino Nano 2.x). It has more or less the same functionality of the Arduino Duemilanove, but in a different package. It lacks only a DC power jack, and works with a Mini-B USB cable instead of a standard one. The Nano was designed and is being produced by Gravitech.

Schematic and Design

Arduino Nano 3.0 (ATmega328): schematic, Eagle files.
Arduino Nano 2.3 (ATmega168): manual (pdf), Eagle files. *Note:* since the free version of Eagle does not handle more than 2 layers, and this version of the Nano is 4 layers, it is published here unrouted, so users can open and use it in the free version of Eagle.

Specifications:

Microcontroller	Atmel ATmega168 or ATmega328
Operating Voltage (logic level)	5 V
Input Voltage (recommended)	7-12 V
Input Voltage (limits)	6-20 V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	8
DC Current per I/O Pin	40 mA
Flash Memory	16 KB (ATmega168) or 32 KB (ATmega328) of which 2 KB used by bootloader
SRAM	1 KB (ATmega168) or 2 KB (ATmega328)
EEPROM	512 bytes (ATmega168) or 1 KB (ATmega328)
Clock Speed	16 MHz
Dimensions	0.73" x 1.70"

Power:

The Arduino Nano can be powered via the Mini-B USB connection, 6-20V unregulated external power supply (pin 30), or 5V regulated external power supply (pin 27). The power source is automatically selected to the highest voltage source.

The FTDI FT232RL chip on the Nano is only powered if the board is being powered over USB. As a result, when running on external (non-USB) power, the 3.3V output (which is supplied by the FTDI chip) is not available and the RX and TX LEDs will flicker if digital pins 0 or 1 are high.

Memory

The ATmega168 has 16 KB of flash memory for storing code (of which 2 KB is used for the bootloader); the ATmega328 has 32 KB, (also with 2 KB used for the bootloader). The ATmega168 has 1 KB of SRAM and 512 bytes of EEPROM (which can be read and written with the EEPROM library); the ATmega328 has 2 KB of SRAM and 1 KB of EEPROM.

Input and Output

Each of the 14 digital pins on the Nano can be used as an input or output, using pinMode(), digitalWrite(), and digitalRead() functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kOhms. In addition, some pins have specialized functions:

- **Serial: 0 (RX) and 1 (TX).** Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the FTDI USB-to-TTL Serial chip.
- **External Interrupts: 2 and 3.** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attachInterrupt() function for details.
- **PWM: 3, 5, 6, 9, 10, and 11.** Provide 8-bit PWM output with the analogWrite() function.
- **SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK).** These pins support SPI communication, which, although provided by the underlying hardware, is not currently included in the Arduino language.
- **LED: 13.** There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

The Nano has 8 analog inputs, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though is it possible to change the upper end of their range using the analogReference() function. Additionally, some pins have specialized functionality:

I²C: 4 (SDA) and 5 (SCL). Support I²C (TWI) communication using the Wire library (documentation on the Wiring website).

There are a couple of other pins on the board:

AREF. Reference voltage for the analog inputs. Used with analogReference().
Reset. Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

See also the mapping between Arduino pins and ATmega168 ports.

Communication

The Arduino Nano has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega168 and ATmega328 provide UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An FTDI FT232RL on the board channels this serial communication over USB and the FTDI drivers (included with the Arduino software) provide a virtual com port to software on the computer. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the FTDI chip and USB connection to the computer (but not for serial communication on pins 0 and 1).
A SoftwareSerial library allows for serial communication on any of the Nano's digital pins.
The ATmega168 and ATmega328 also support I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus; see the documentation for details. To use the SPI communication, please see the ATmega168 or ATmega328 datasheet.

Programming

The Arduino Nano can be programmed with the Arduino software (download). Select "Arduino Diecimila, Duemilanove, or Nano w/ ATmega168" or "Arduino Duemilanove or Nano w/ ATmega328" from the **Tools**

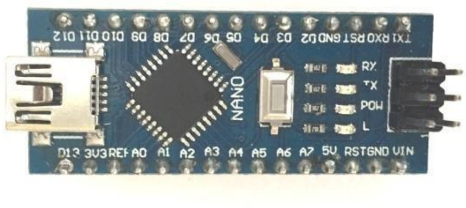
> **Board** menu (according to the microcontroller on your board). For details, see the reference and tutorials.
The ATmega168 or ATmega328 on the Arduino Nano comes preburned with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol (reference, C header files).
You can also bypass the bootloader and program the microcontroller through the ICSP (In-Circuit Serial Programming) header; see these instructions for details.

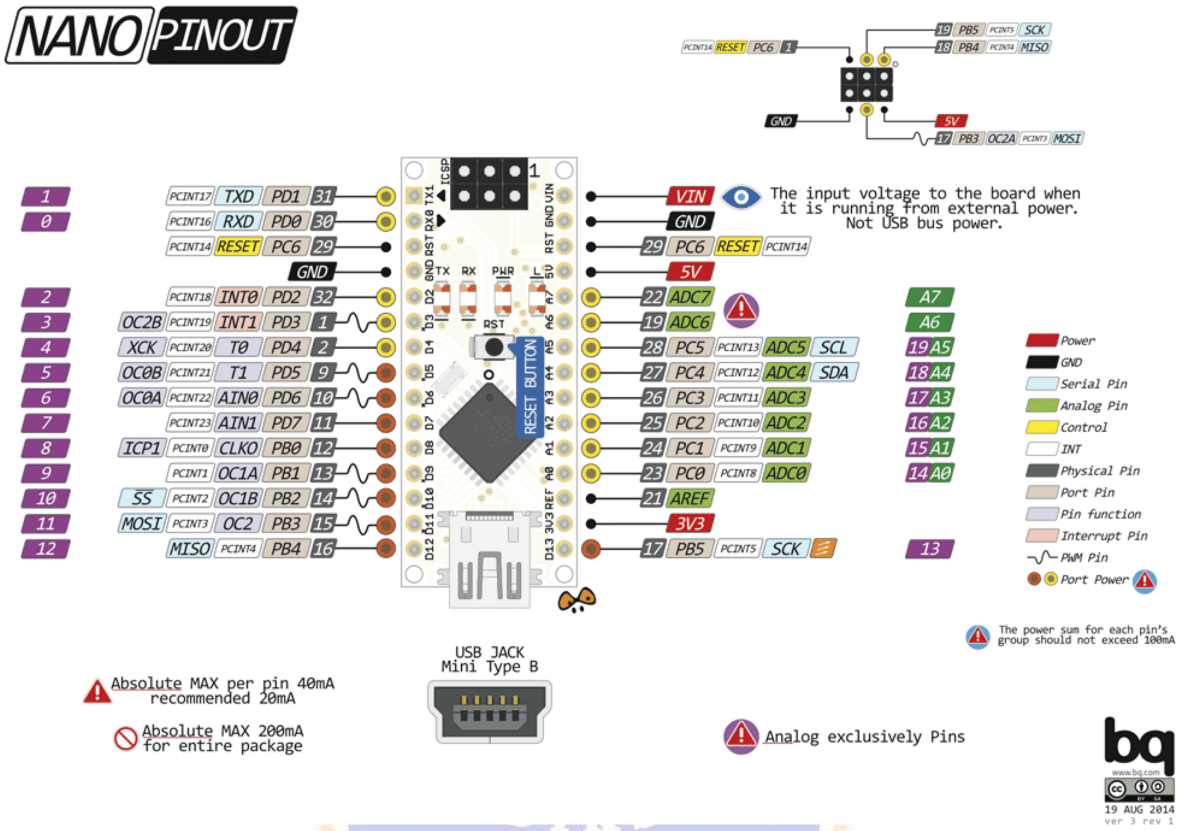
Automatic (Software) Reset

Rather then requiring a physical press of the reset button before an upload, the Arduino Nano is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the FT232RL is connected to the reset line of the ATmega168 or ATmega328 via a 100 nanofarad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip. The Arduino software uses this capability to allow you to upload code by simply pressing the upload button in the Arduino environment. This means that the bootloader can have a shorter timeout, as the lowering of DTR can be well-coordinated with the start of the upload.
This setup has other implications. When the Nano is connected to either a computer running Mac OS X or Linux, it resets each time a connection is made to it from software (via USB). For the following half-second or so, the bootloader is running on the Nano. While it is programmed to ignore malformed data (i.e. anything besides an upload of new code), it will intercept the first few bytes of data sent to the board after a connection is opened. If a sketch running on the board receives one-time configuration or other data when it first starts, make sure that the software with which it communicates waits a second after opening the connection and before sending this data.

LAMPIRAN

Datasheet Arduino Nano





Arduino Nano Pin Configuration

Pin Category	Pin Name	Details
Power	**Vin, 3.3V, 5V, GND**	**Vin:** Input voltage to Arduino when using an external power 12V). **5V:** Regulated power supply used to power microcontroller components on the board. **3.3V:** 3.3V supply generated by on-board voltage regulator. current draw is 50mA. **GND:** Ground pins.

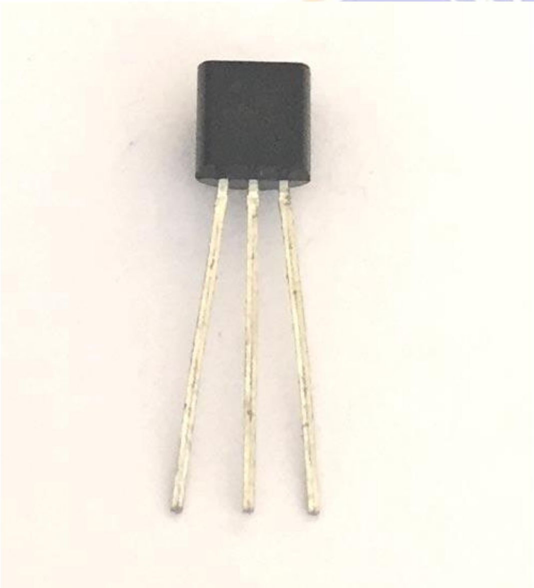
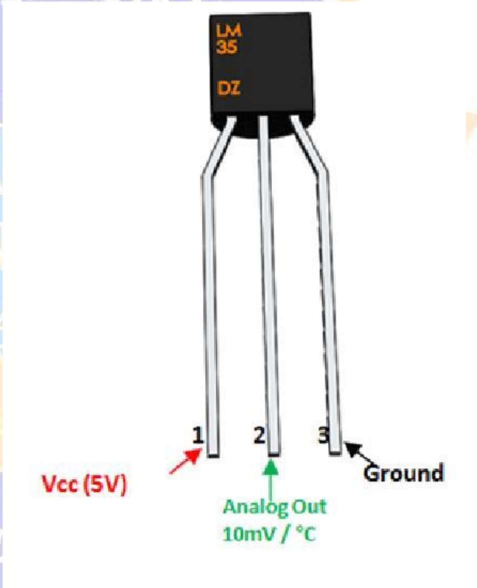
Reset	**Reset**	Resets the microcontroller.
Analog Pins	**A0 – A7**	Used to measure analog voltage in the range of 0-5V
Input/Output Pins	**Digital Pins D0 - D13**	Can be used as input or output pins. 0V (low) and 5V (high)
Serial	Rx, **Tx**	Used to receive and transmit TTL serial data.
External Interrupts	2, 3	To trigger an interrupt.
PWM	3, 5, 6, 9, 11	Provides 8-bit PWM output.
SPI	10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK)	Used for SPI communication.
Inbuilt LED	**13**	To turn on the inbuilt LED.
IIC	A4 (SDA), A5 (SCA)	Used for TWI communication.
AREF	**AREF**	To provide reference voltage for input voltage.

Arduino Nano Technical Specifications

Microcontroller	ATmega328P – 8 bit AVR family microcontroller
Operating Voltage	5V
Recommended Input Voltage for Vin pin	7-12V
Analog Input Pins	6 (A0 – A5)
Digital I/O Pins	14 (Out of which 6 provide PWM output)
DC Current on I/O Pins	40 mA

DC Current on 3.3V Pin	50 mA
Flash Memory	32 KB (2 KB is used for Bootloader)
SRAM	2 KB
EEPROM	1 KB
Frequency (Clock Speed)	16 MHz
Communication	IIC, SPI, USART

Sensor Suhu

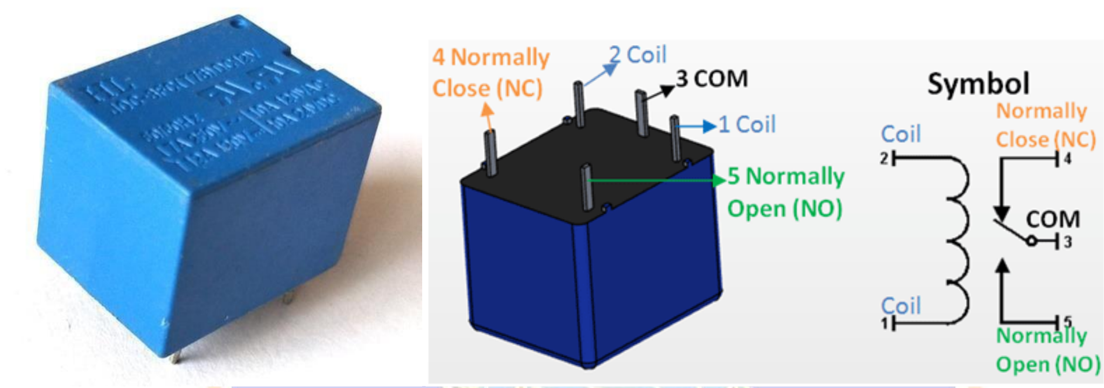
Pin Configuration:

Pin Number	**Pin Name**	**Description**
1	Vcc	Input voltage is +5V for typical applications
2	Analog Out	There will be increase in 10mV for raise of every 1°C. Can range from -1V 6V(150°C)
3	Ground	Connected to ground of circuit

LM35 Regulator Features:

- Minimum and Maximum Input Voltage is 35V and -2V respectively. Typically 5V.
- Can measure temperature ranging from -55°C to 150°C
- Output voltage is directly proportional (Linear) to temperature (i.e.) there will be a rise of 10mV (0.01V) for every 1°C rise in temperature.
- ±0.5°C Accuracy
- Drain current is less than 60uA
- Low cost temperature sensor
- Small and hence suitable for remote applications
- Available in TO-92, TO-220, TO-CAN and SOIC package

Modul Rellay



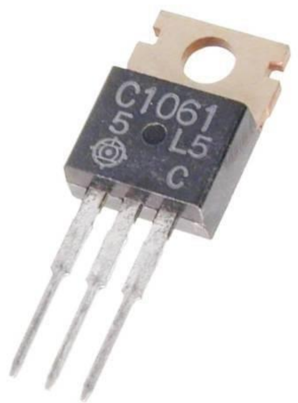
Relay Pin Configuration

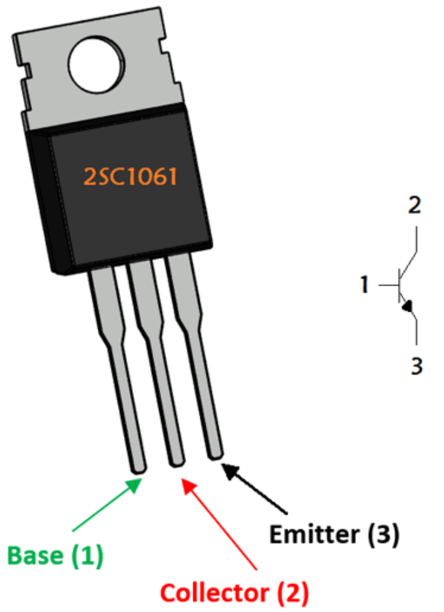
Pin Number	Pin Name	Description
1	Coil End 1	Used to trigger(On/Off) the Relay, Normally one end is connected to 5V and the other end to ground
2	Coil End 2	Used to trigger(On/Off) the Relay, Normally one end is connected to 5V and the other end to ground
3	Common (COM)	Common is connected to one End of the Load that is to be controlled
4	Normally Close (NC)	The other end of the load is either connected to NO or NC. If connected to NC the load remains connected before trigger
5	Normally Open (NO)	The other end of the load is either connected to NO or NC. If

		connected to NO the load remains disconnected before trigger

Features of 5-Pin 5V Relay

- Trigger Voltage (Voltage across coil) : 5V DC
- Trigger Current (Nominal current) : 70mA
- Maximum AC load current: 10A @ 250/125V AC
- Maximum DC load current: 10A @ 30/28V DC
- Compact 5-pin configuration with plastic moulding
- Operating time: 10msec Release time: 5msec
- Maximum switching: 300 operating/minute (mechanically)





Pin Description

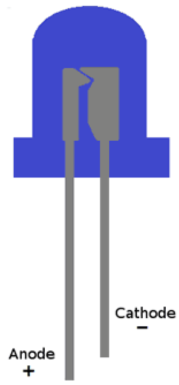
Pin Number	Pin Name	Description
1	Base	Controls the biasing of transistor, Used to turn ON or OFF the transistor
2	Collector	Current flows in through collector, normally connected to load
3	Emitter	Current Drains out through emitter, normally connected to ground

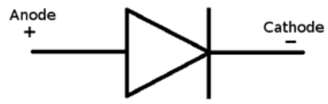
Features

- NPN Power Transistor

- Low Frequency High Power amplifier
- Continuous Collector current (IC) is 3A
- Collector-Emitter voltage (VCE) is 50 V
- Collector-Base voltage (VCB) is 50
- Emitter Base Voltage (VBE) is 4V
- DC Current Gain (hfe): 35 to 320
- Available in To-220 Package

LED





Pin Configuration

Pin Name	**Description**
Anode	Positive terminal of LED
Cathode	Negative terminal of LED

Features and Technical Specifications

- Superior weather resistance
- 5mm Round Standard Directivity
- UV Resistant Eproxy
- Forward Current (IF): 30mA
- Forward Voltage (VF): 1.8V to 2.4V
- Reverse Voltage: 5V
- Operating Temperature: -30℃ to +85℃
- Storage Temperature: -40℃ to +100℃
- Luminous Intensity: 20mcd

Kabel Jumper

Male toFemale

Kabel yang di gunakan pada rangkaian egg boiler berbasis arduino yang paling sering digunakan yaitu kabel male to famale, karena sebagai penghubung sangat bagus untuk di kombinasikan dengan rangkaian egg boiler berbasis arduino. Kabel male to famale memiliki dua header yang berbeda itu menjadikan kabel ini biasanya disebut kabel male to famale.

DIODA

VISHAY.

1N4001 thru 1N4007

Vishay General Semiconductor

General Purpose Plastic Rectifier

DO-204AL (DO-41)

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	1.0 A
V_{RRM}	50 V to 1000 V
I_{FSM} (8.3 ms sine-wave)	30 A
I_{FSM} (square wave t_p = 1 ms)	45 A
V_F	1.1 V
I_R	5.0 µA
T_J max.	150 °C

FEATURES

- Low forward voltage drop
- Low leakage current
- High forward surge capability
- Solder dip 275 °C max. 10 s, per JESD 22-B106
- Compliant to RoHS Directive 2002/95/EC and in accordance to WEEE 2002/96/EC

Pb e3 RoHS COMPLIANT

TYPICAL APPLICATIONS

For use in general purpose rectification of power supplies, inverters, converters and freewheeling diodes application.

Note

- These devices are not AEC-Q101 qualified.

MECHANICAL DATA

Case: DO-204AL, molded epoxy body
Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS compliant, commercial grade

Terminals: Matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 suffix meets JESD 201 class 1A whisker test

Polarity: Color band denotes cathode end

RESISTOR

EM-18 RFID Reader

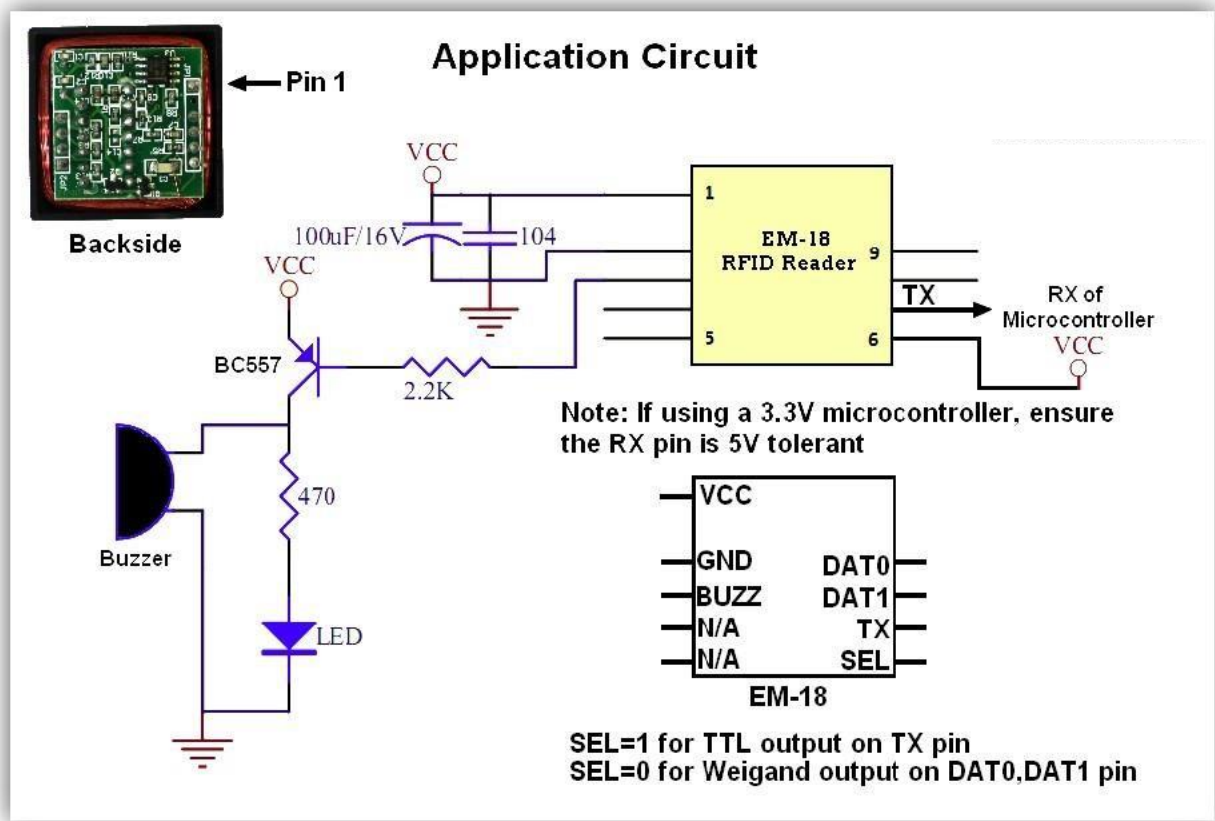
The EM-18 RFID Reader module operating at 125kHz is an inexpensive solution for your RFID based application. The Reader module comes with an on-chip antenna and can be powered up with a 5V power supply. Power-up the module and connect the transmit pin of the module to recieve pin of your microcontroller. Show your card within the reading distance and the card number is thrown at the output. Optionally the module can be configured for also a weigand output.

Typical Applications

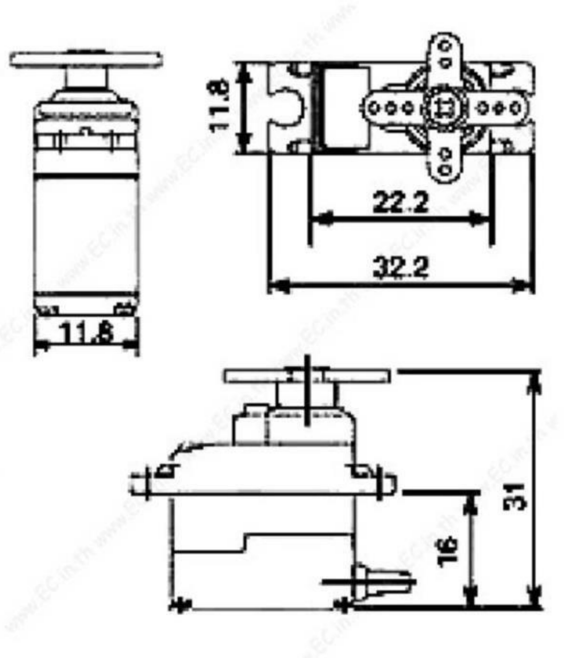
- e-Payment
- e-Toll Road Pricing
- e-Ticketing for Events
- e-Ticketing for Public Transport
- Access Control
- PC Access
- Authentication
- Printer / Production Equipment

Features

RF Transmit Frequency	125kHz
Supported Standards	EM4001 64-bit RFID tag compatible
Communications Interface	TTL Serial Interface, Wiegand output
Communications Protocol	Specific ASCII
Communications Parameter	9600 bps, 8, N, 1
Power Supply	4.6V - 5.5VDC ± 10% regulated
Current Consumption	50 mA < 10mA at power down mode.
Reading distance	Up to 100mm, depending on tag
Antenna	Integrated
Size (LxWxH)	32 x 32 x 8mm



SG90 9 g Micro Servo



Tiny and lightweight with high output power. Servo can rotate approximately 180 degrees (90 in each direction), and works just like the standard kinds but *smaller*. You can use any servo code, hardware or library to control these servos. Good for beginners who want to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places. It comes with a 3 horns (arms) and hardware.

Specifications

- Weight: 9 g
- Dimension: 22.2 x 11.8 x 31 mm approx.
- Stall torque: 1.8 kgf·cm
- Operating speed: 0.1 s/60 degree
- Operating voltage: 4.8 V (~5V)
- Dead band width: 10 µs
- Temperature range: 0 ºC – 55 ºC

Position "0" (1.5 ms pulse) is middle, "90" (~2ms pulse) is all the way to the left. ms pulse) is all the way to the right, ""-90" (~1ms pulse) is all the way to the left.

TowerPro SG90 - Micro Servo

Basic Information

Modulation: Analog

Torque: **4.8V:** 25.0 oz-in (1.80 kg-cm)

Speed: **4.8V:** 0.10 sec/60°

Weight: 0.32 oz (9.0 g)

Dimensions:

Length: 0.91 in (23.1 mm)

Width: 0.48 in (12.2 mm)

Height:1.14 in (29.0 mm)

Motor Type: 3-pole
Gear Type: Plastic
Rotation/Support: Bushing

Additional Specifications

Rotational Range:180°
Pulse Cycle: ca. 20 ms
Pulse Width: 500-2400 µs

T-Pro Mini Servo SG-90 9G Servo

The **TP SG90** is similar in size and weight to the Hitec HS-55, and is a good choice for most park flyers and helicopters. Hobbyists from around the world has used the SG90 on famous planes like GWS Slow Stick, E-Flite Airplanes, Great Planes, Thunder Tiger, Align, EDF jets and more. If you are looking for a servo that won't break your arm or leg, this is the perfect choice for you!

The **TP SG90** servo weighs 0.32 ounces (9.0 grams). Total weight with wire and connector is 0.37 ounces (10.6 grams).

The **TP SG90** has the universal "S" type connector that fits most receivers, including Futaba, JR, GWS, Cirrus, Blue Bird, Blue Arrow, Corona, Berg and Hitec.

The wire colors are **Red = Battery(+)** **Brown = Battery(-)** **Orange = Signal**

TP SG90 Specifications:

Dimensions (L x W x H) = 0.86 x 0.45 x 1.0 inch (22.0 x 11.5 x 27 mm)

Weight = 0.32 ounces (9 grams)

Weight with wire and connecter = 0.37 ounce (10.6 grams)

Stall Torque at 4.8 volts = 16.7 oz/in (1.2 kg/cm)

Operating Voltage = 4.0 to 7.2 volts

Operating Speed at 4.8 volts (no load) = 0.12 sec/ 60 degrees

Connector Wire Length = 9.75 inches (248 mm)

The TG9e boasts the same performance as other servo's 10x the price with a .10sec travel time and up to 1.5kg in torque and an ultra narrow dead bandwidth!

The TG9e performance is on par with the famous HXT900, however the TG9e isn't as resistant to crashes or over-loading.

Please always ensure your control surfaces are bind free.

Spec.

Dimension: 23x12.2x29mm

Torque: 1.5kg/cm (4.8V)

Operating speed: 0.10sec/60 degree 0.09sec/60 degree(6.0V)

Operating voltage: 4.8V

Temperature range: 0-55C

Dead band-width: 7us

Lead Length: 260mm